

GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

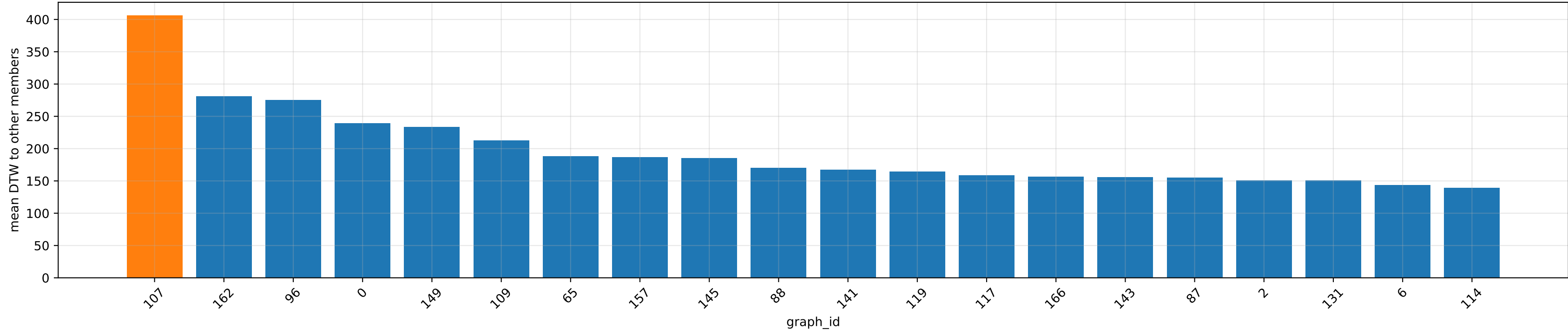
First applied split summary | seed=0

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
0	17	0	1 vs 19	107	107	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
406.254	nan	172.658

Per-member mean DTW



GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

First applied split pairwise DTW metadata | seed=0

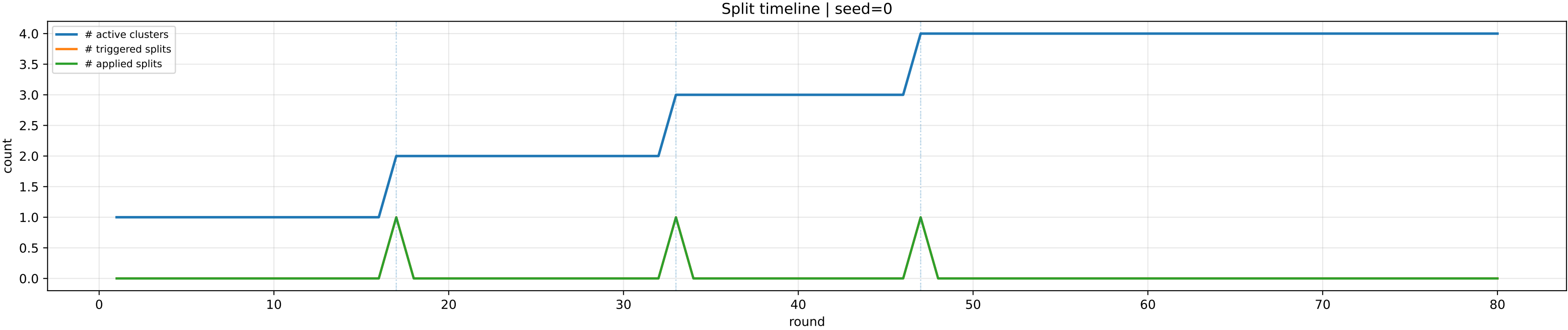
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
0	17	0	20	107.0	0.051	406.254	1

Pairwise DTW table | first applied split | seed=0
Global max DTW = 540.9 at (145, 107)

graph_id	87	157	149	109	143	114	117	2	88	145	166	141	65	6	131	119	96	162	107	0
87	0.0	77.9	264.1	104.8	145.5	58.0	140.0	113.6	137.1	200.2	66.7	114.9	173.3	94.3	119.3	150.9	219.8	267.8	340.8	157.6
157	77.9	0.0	204.5	122.9	98.1	145.6	208.9	203.8	206.4	236.6	147.8	136.4	243.0	186.6	166.8	240.8	216.3	237.1	297.9	164.8
149	264.1	204.5	0.0	344.4	133.7	196.9	165.1	186.1	209.3	166.0	290.5	191.2	205.0	199.1	171.7	173.3	290.8	285.8	456.1	306.5
109	104.8	122.9	344.4	0.0	244.0	165.7	258.9	224.7	263.3	308.1	139.6	161.7	308.1	157.4	239.2	278.9	127.8	177.8	239.0	178.6
143	145.5	98.1	133.7	244.0	0.0	111.1	57.7	97.7	98.9	106.2	158.2	101.5	115.1	92.0	73.6	79.7	296.6	279.8	443.5	226.7
114	58.0	145.6	196.9	165.7	111.1	0.0	68.1	60.2	91.7	113.0	78.0	81.9	118.1	69.4	79.4	85.0	239.8	244.7	422.8	210.9
117	140.0	208.9	165.1	258.9	57.7	68.1	0.0	59.7	81.7	61.7	126.9	115.7	79.6	74.6	42.7	43.2	339.6	322.2	491.7	274.9
2	113.6	203.8	186.1	224.7	97.7	60.2	59.7	0.0	83.8	61.9	82.6	138.4	79.5	36.2	50.1	62.0	306.5	297.5	470.5	253.7
88	137.1	206.4	209.3	263.3	98.9	91.7	81.7	83.8	0.0	58.5	100.4	114.0	58.1	114.8	69.1	68.1	337.4	336.4	509.2	290.3
145	200.2	236.6	166.0	308.1	106.2	113.0	61.7	61.9	58.5	0.0	162.3	141.1	72.6	108.2	73.9	38.4	394.0	354.6	540.9	324.1
166	66.7	147.8	290.5	139.6	158.2	78.0	126.9	82.6	100.4	162.3	0.0	159.0	130.7	54.3	110.2	99.7	232.9	281.9	379.9	170.6
141	114.9	136.4	191.2	161.7	101.5	81.9	115.7	138.4	114.0	141.1	159.0	0.0	127.8	150.8	137.1	137.1	243.5	241.8	425.7	259.2
65	173.3	243.0	205.0	308.1	115.1	118.1	79.6	79.5	58.1	72.6	130.7	127.8	0.0	92.3	98.4	68.6	382.5	356.8	540.9	324.1
6	94.3	186.6	199.1	157.4	92.0	69.4	74.6	36.2	114.8	108.2	54.3	150.8	92.3	0.0	57.0	65.5	252.1	277.9	432.8	207.6
131	119.3	166.8	171.7	239.2	73.6	79.4	42.7	50.1	69.1	73.9	110.2	137.1	98.4	57.0	0.0	49.5	328.3	289.3	462.3	245.4
119	150.9	240.8	173.3	278.9	79.7	85.0	43.2	62.0	68.1	38.4	99.7	137.1	68.6	65.5	49.5	0.0	355.4	327.0	511.8	294.9
96	219.8	216.3	290.8	127.8	296.6	239.8	339.6	306.5	337.4	394.0	232.9	243.5	382.5	252.1	328.3	355.4	0.0	201.1	238.6	233.0
162	267.8	237.1	285.8	177.8	279.8	244.7	322.2	297.5	336.4	354.6	281.9	241.8	356.8	277.9	289.3	327.0	201.1	0.0	327.0	233.8
107	340.8	297.9	456.1	239.0	443.5	422.8	491.7	470.5	509.2	540.9	379.9	425.7	540.9	432.8	462.3	511.8	238.6	327.0	0.0	187.5
0	157.6	164.8	306.5	178.6	226.7	210.9	274.9	253.7	290.3	324.1	170.6	259.2	324.1	207.6	245.4	294.9	233.0	233.8	187.5	0.0

GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | split
timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

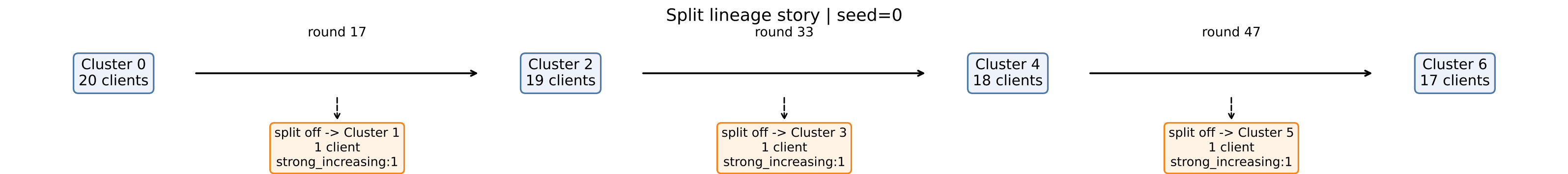


Seed summary | seed=0

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
0	0.25	0.7	5.0	5.0	2.0	1.0	17, 33, 47	17, 33, 47	1, 3, 5, 6	1:1, 3:1, 5:1, 6:17

GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095



Applied split story | seed=0

round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
17	0	20	166.133 < 166.917 , 341.348 > 311.554	C1 (1 client)	strong_increasing:1	C2 (19 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:3
33	2	19	162.466 < 162.528 , 311.238 > 306.820	C3 (1 client)	strong_increasing:1	C4 (18 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:2
47	4	18	160.775 < 160.922 , 291.150 > 280.818	C5 (1 client)	strong_increasing:1	C6 (17 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:1

GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | final active clusters

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

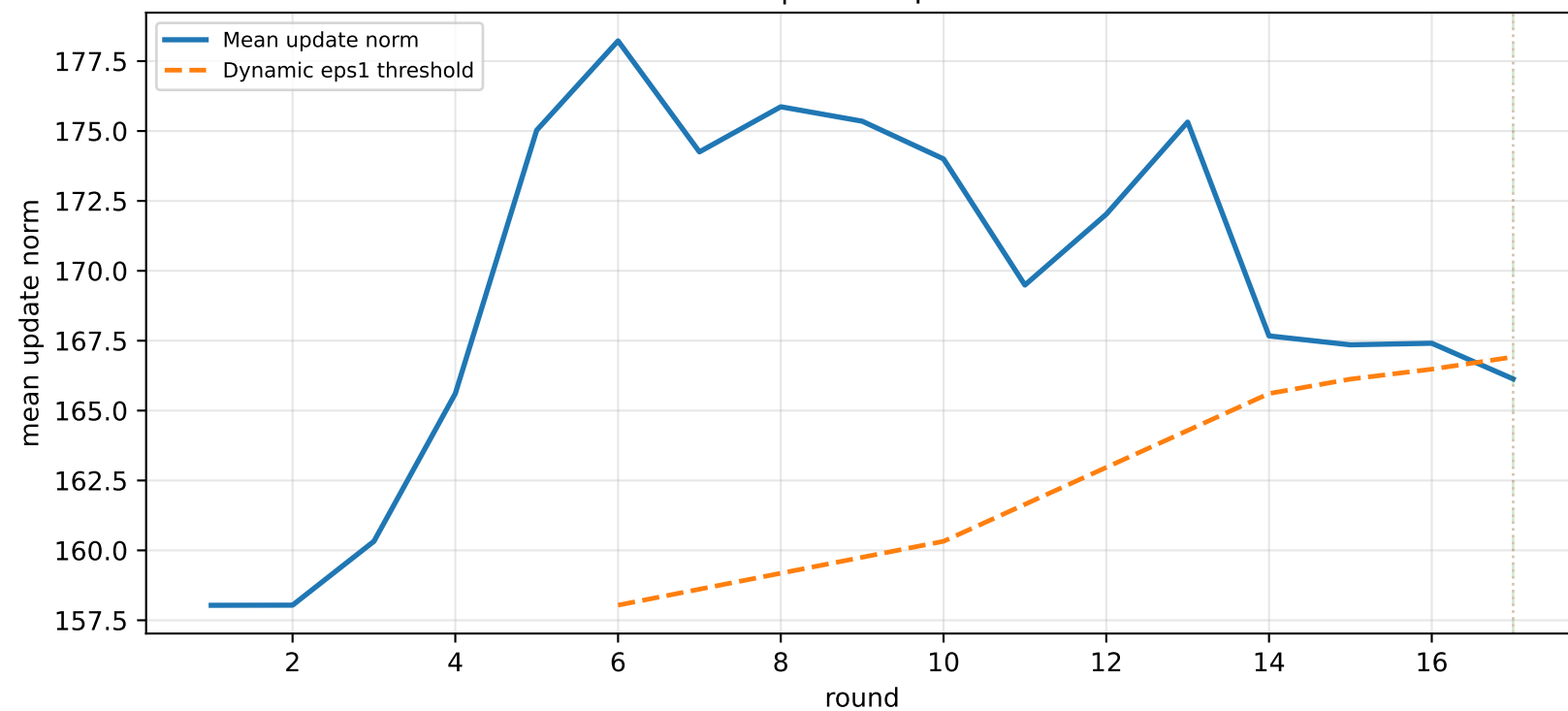
Final active clusters | seed=0

cluster_id	cluster_size	members	member_indices	family_mix
1	1	107	18	strong_increasing:1
3	1	96	16	strong_increasing:1
5	1	162	17	strong_increasing:1
6	17	87, 157, 149, 109, 143, 114, 117, 2, 88, 145, 166, 141, 65, 6, 131, 119, 0	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 19	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:1

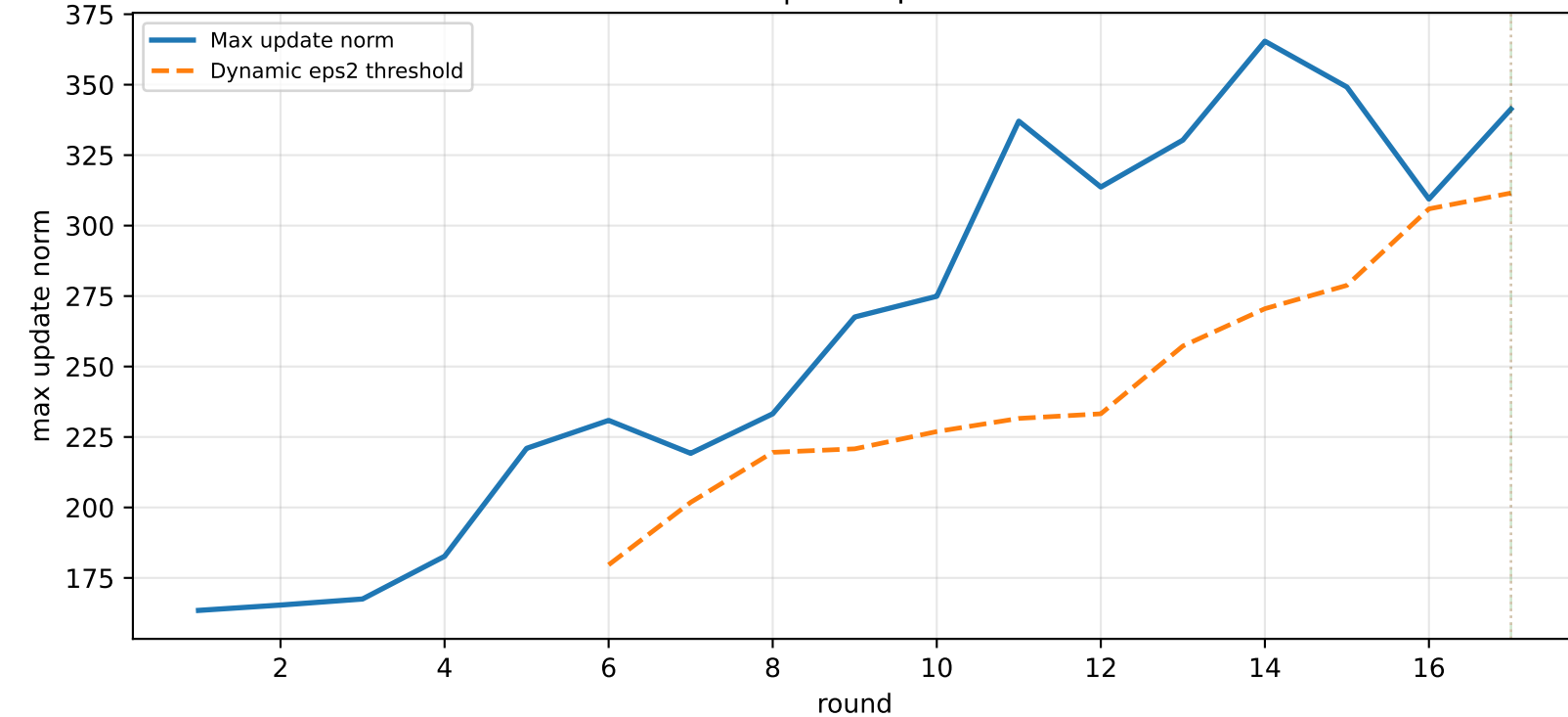
GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

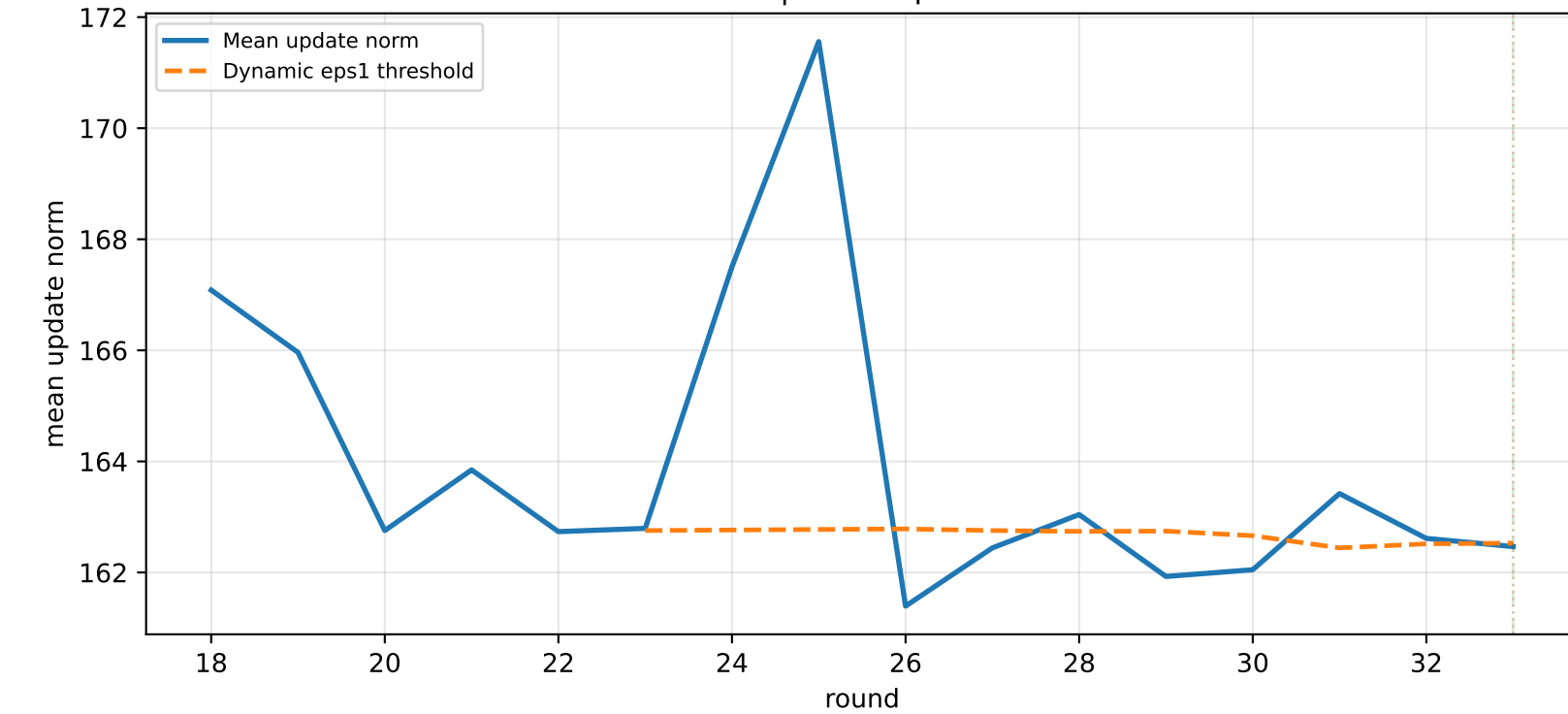
Cluster 0 | Mean update norm



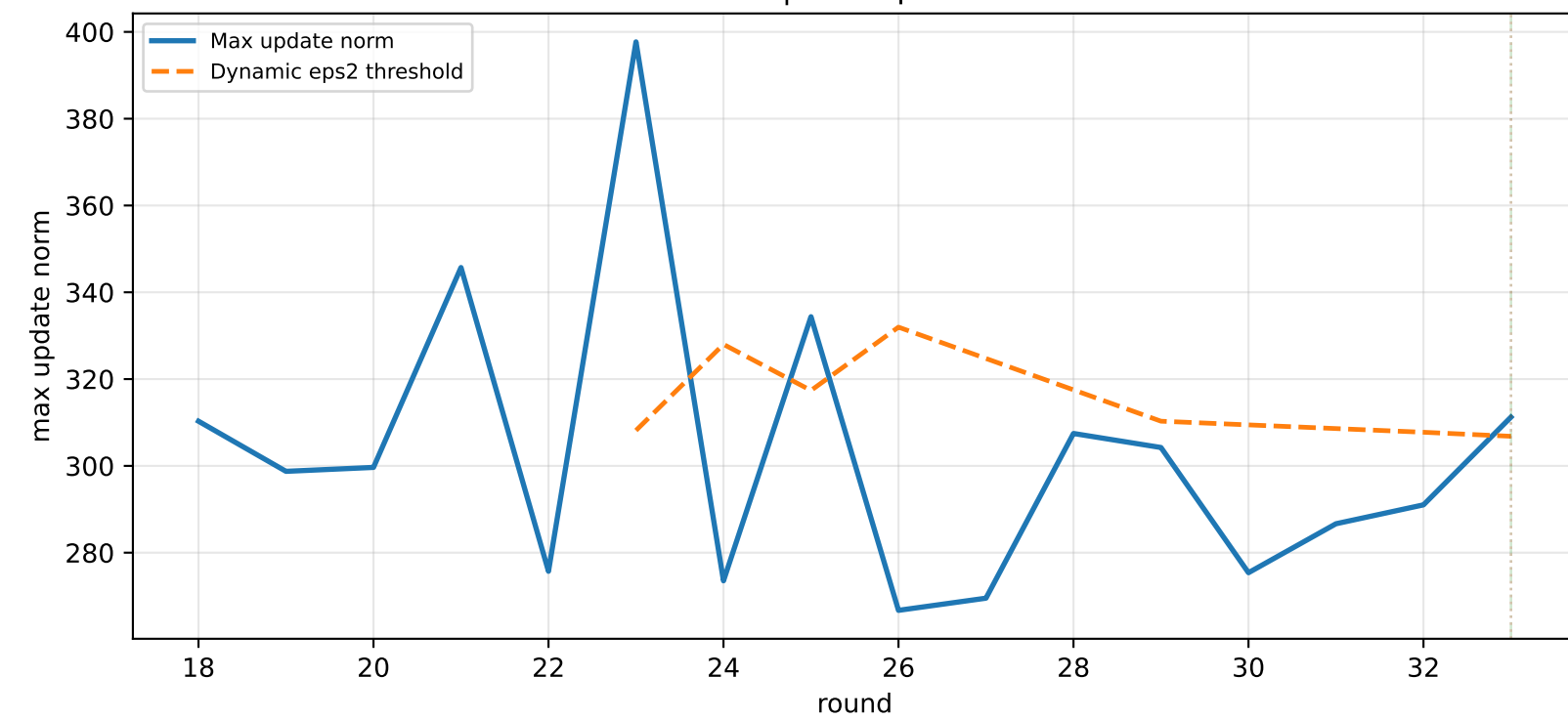
Cluster 0 | Max update norm



Cluster 2 | Mean update norm



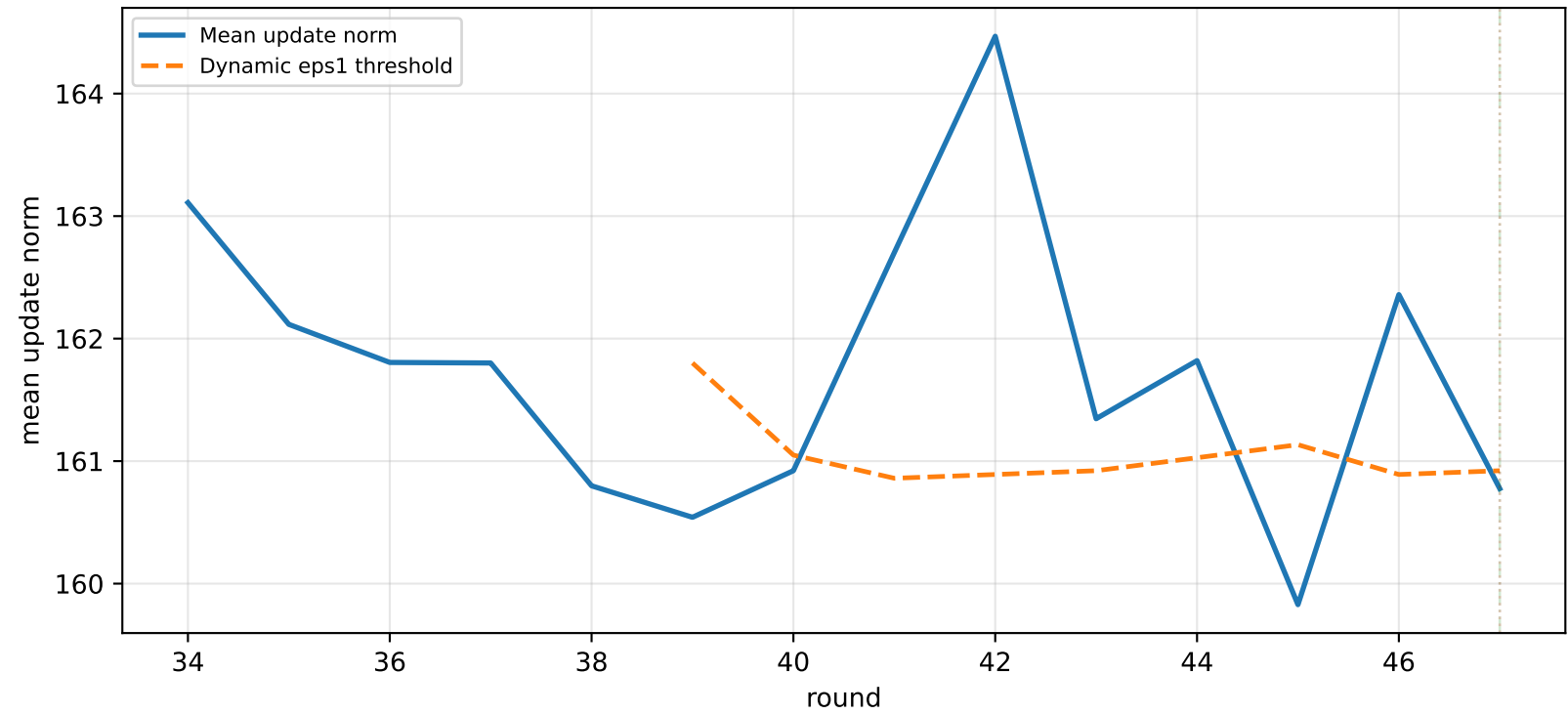
Cluster 2 | Max update norm



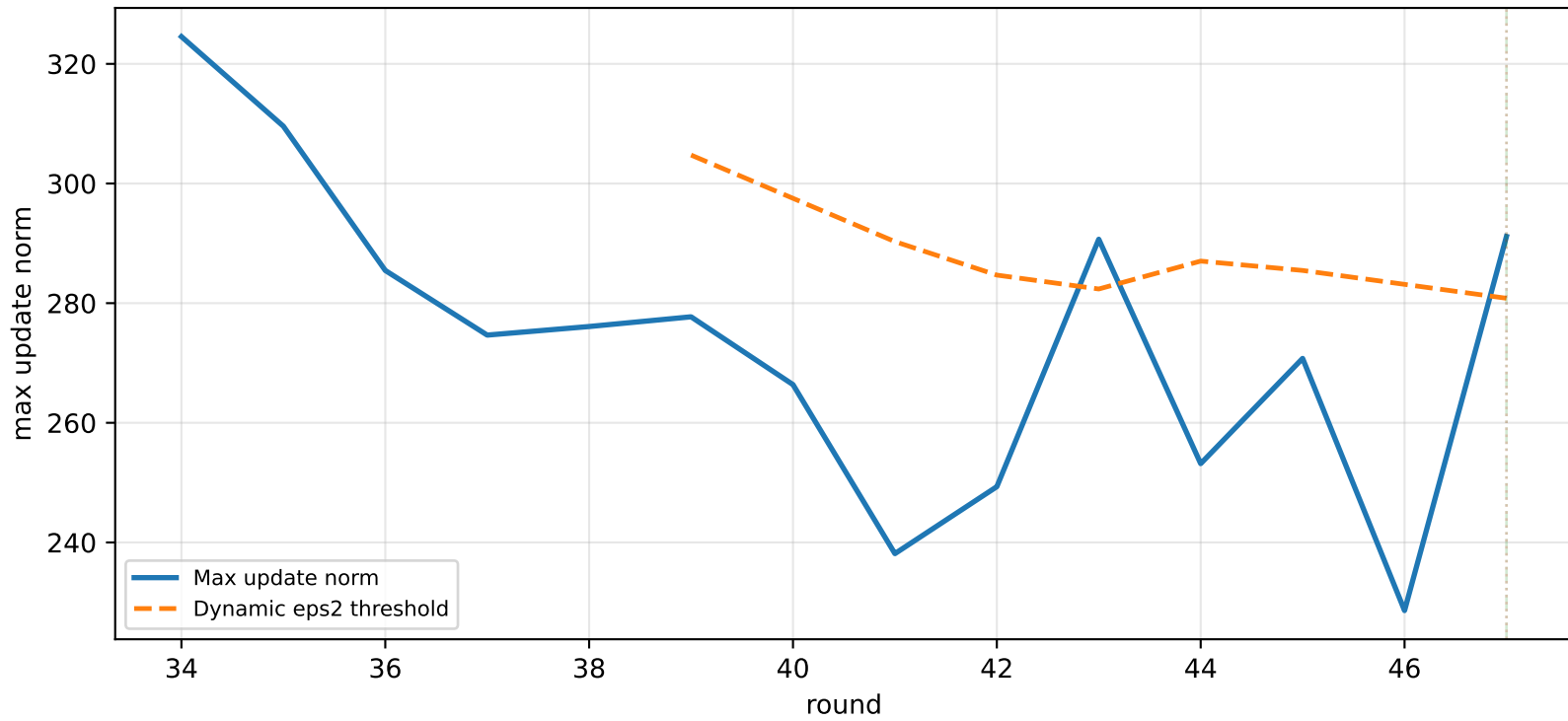
GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=0 | norm / threshold panels (4-6)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

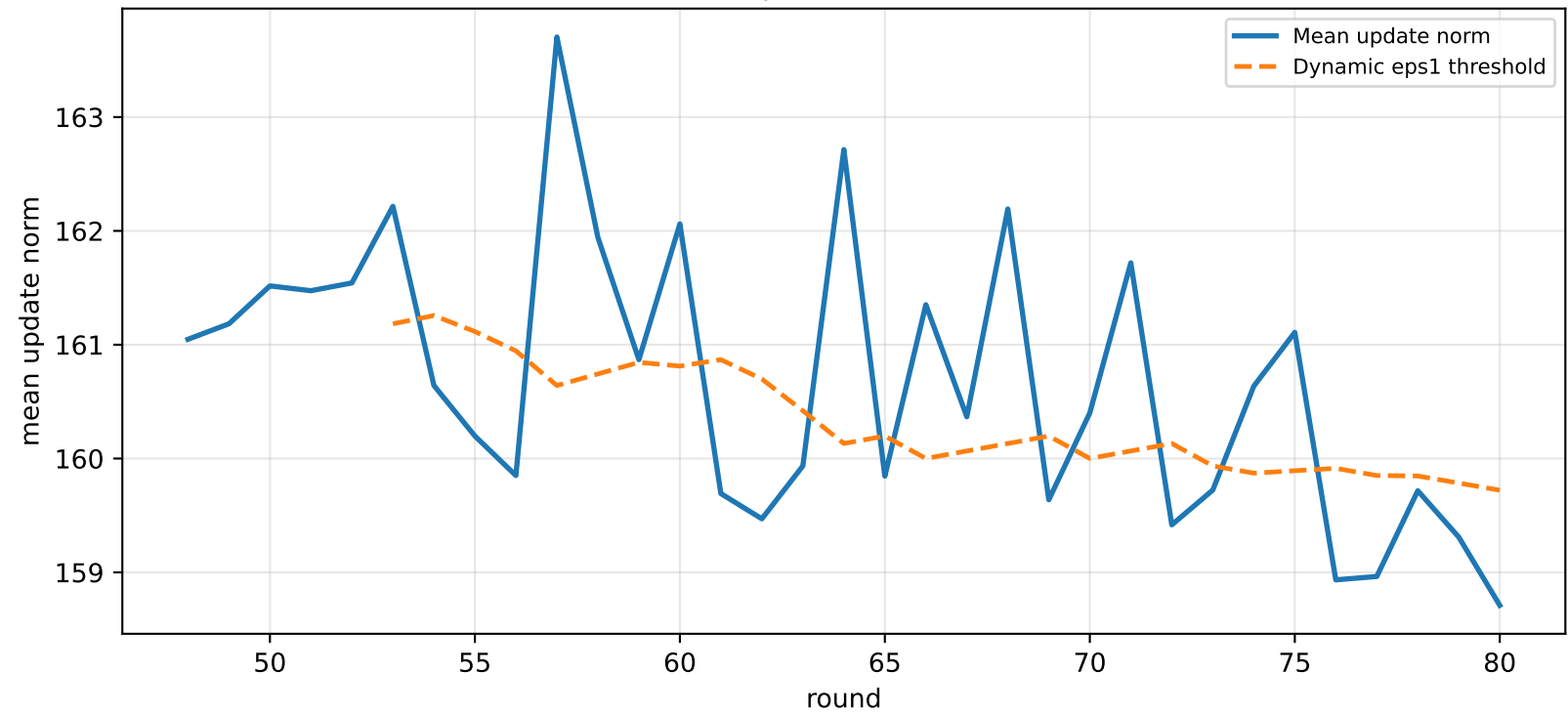
Cluster 4 | Mean update norm



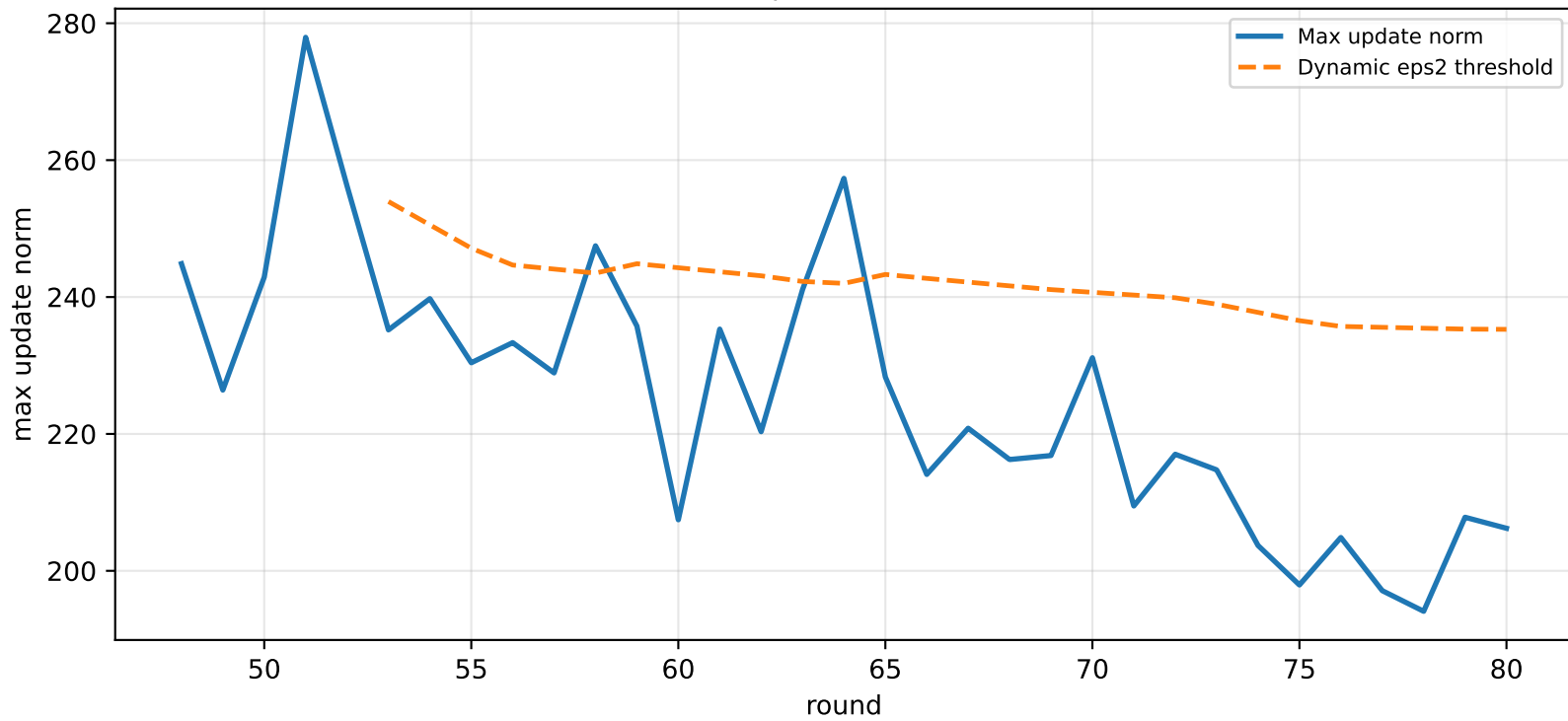
Cluster 4 | Max update norm



Cluster 6 | Mean update norm



Cluster 6 | Max update norm



GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

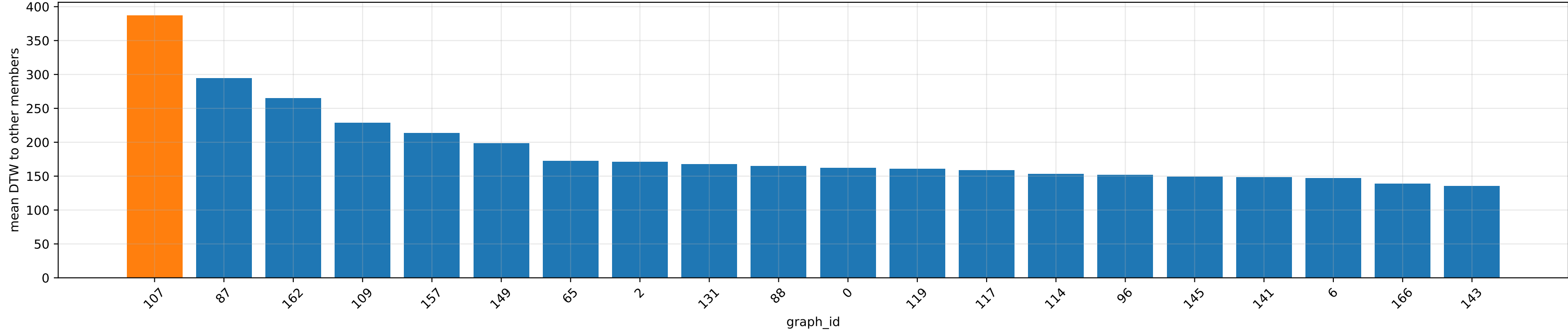
First applied split summary | seed=1

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
1	15	0	1 vs 19	107	107	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
387.101	nan	166.390

Per-member mean DTW



GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

First applied split pairwise DTW metadata | seed=1

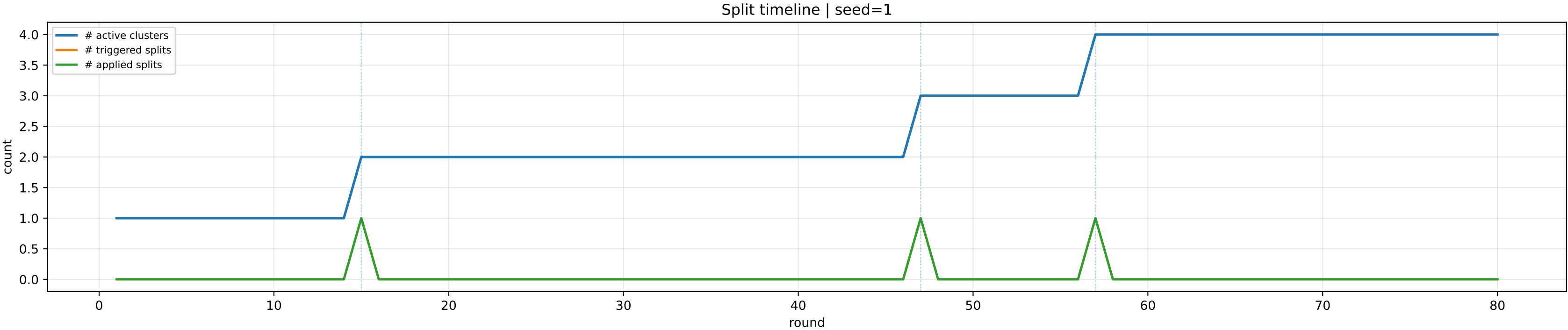
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
1	15	0	20	107.0	0.051	387.101	1

Pairwise DTW table | first applied split | seed=1
Global max DTW = 464.0 at (88, 107)

graph_id	87	157	149	109	143	114	117	2	88	145	166	141	65	6	131	119	96	162	107	0
87	0.0	195.7	189.7	181.8	312.9	217.4	399.3	315.9	409.6	378.0	304.0	345.8	384.9	325.7	405.4	394.1	256.9	150.8	197.8	234.9
157	195.7	0.0	82.5	190.8	209.9	203.2	235.0	239.7	226.1	233.5	214.9	211.4	250.8	239.5	265.8	219.9	155.5	137.9	402.5	148.6
149	189.7	82.5	0.0	218.0	179.7	138.4	231.2	227.0	230.0	207.0	177.0	168.6	219.8	217.9	244.8	235.8	180.4	161.7	338.5	128.4
109	181.8	190.8	218.0	0.0	189.8	141.0	273.1	212.5	237.3	236.5	204.7	236.5	289.0	228.3	290.7	267.2	204.6	176.7	409.8	158.2
143	312.9	209.9	179.7	189.8	0.0	89.3	73.7	86.2	98.6	61.6	71.2	69.7	79.0	38.2	90.9	87.6	106.9	267.1	375.7	85.3
114	217.4	203.2	138.4	141.0	89.3	0.0	132.2	128.8	123.2	112.6	85.9	117.0	118.1	105.8	128.8	143.8	154.6	239.3	351.3	185.0
117	399.3	235.0	231.2	273.1	73.7	132.2	0.0	156.9	54.4	17.8	62.3	79.2	86.1	75.4	33.4	19.3	127.5	340.8	442.5	169.9
2	315.9	239.7	227.0	212.5	86.2	128.8	156.9	0.0	147.8	130.6	102.2	129.7	44.1	77.7	122.0	173.0	118.4	269.4	428.0	145.2
88	409.6	226.1	230.0	237.3	98.6	123.2	54.4	147.8	0.0	57.3	48.3	86.4	113.4	102.6	76.6	52.6	116.4	325.2	464.0	169.7
145	378.0	233.5	207.0	236.5	61.6	112.6	17.8	130.6	57.3	0.0	59.6	75.1	115.0	71.3	54.3	27.9	117.1	306.1	431.0	141.1
166	304.0	214.9	177.0	204.7	71.2	85.9	62.3	102.2	48.3	59.6	0.0	63.1	96.6	87.6	79.4	68.2	95.6	290.7	390.8	135.3
141	345.8	211.4	168.6	236.5	69.7	117.0	79.2	129.7	86.4	75.1	63.1	0.0	86.5	72.1	82.0	66.5	91.2	301.8	403.5	130.3
65	384.9	250.8	219.8	289.0	79.0	118.1	86.1	44.1	113.4	115.0	96.6	86.5	0.0	58.9	67.9	92.7	150.9	370.3	455.8	193.4
6	325.7	239.5	217.9	228.3	38.2	105.8	75.4	77.7	102.6	71.3	87.6	72.1	58.9	0.0	48.4	94.7	106.2	311.4	397.9	134.5
131	405.4	265.8	244.8	290.7	90.9	128.8	33.4	122.0	76.6	54.3	79.4	82.0	67.9	48.4	0.0	51.0	138.6	358.0	459.7	187.1
119	394.1	219.9	235.8	267.2	87.6	143.8	19.3	173.0	52.6	27.9	68.2	66.5	92.7	94.7	51.0	0.0	119.8	335.1	436.8	164.2
96	256.9	155.5	180.4	204.6	106.9	154.6	127.5	118.4	116.4	117.1	95.6	91.2	150.9	106.2	138.6	119.8	0.0	224.2	317.9	99.6
162	150.8	137.9	161.7	176.7	267.1	239.3	340.8	269.4	325.2	306.1	290.7	301.8	370.3	311.4	358.0	335.1	224.2	0.0	325.9	139.7
107	197.8	402.5	338.5	409.8	375.7	351.3	442.5	428.0	464.0	431.0	390.8	403.5	455.8	397.9	459.7	436.8	317.9	325.9	0.0	325.4
0	234.9	148.6	128.4	158.2	85.3	185.0	169.9	145.2	169.7	141.1	135.3	130.3	193.4	134.5	187.1	164.2	99.6	139.7	325.4	0.0

GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | split
timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

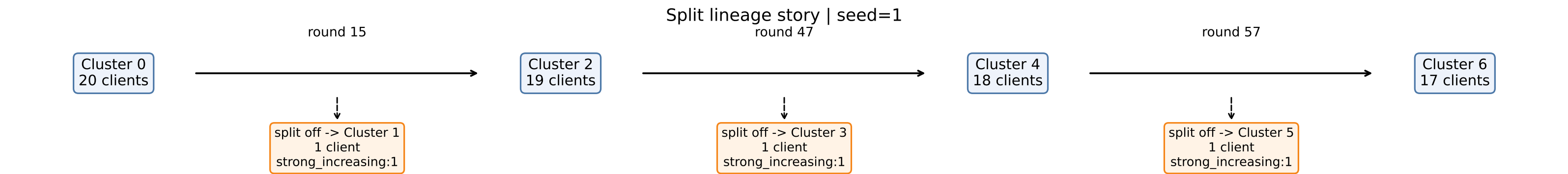


Seed summary | seed=1

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
1	0.25	0.7	5.0	5.0	2.0	1.0	15, 47, 57	15, 47, 57	1, 3, 5, 6	1:1, 3:1, 5:1, 6:17

GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095



Applied split story | seed=1

round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
15	0	20	162.123 < 166.767 , 331.002 > 296.728	C1 (1 client)	strong_increasing:1	C2 (19 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:3
47	2	19	160.030 < 161.578 , 312.293 > 303.061	C3 (1 client)	strong_increasing:1	C4 (18 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:2
57	4	18	159.465 < 160.246 , 271.577 > 268.295	C5 (1 client)	strong_increasing:1	C6 (17 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:1

GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | final active clusters

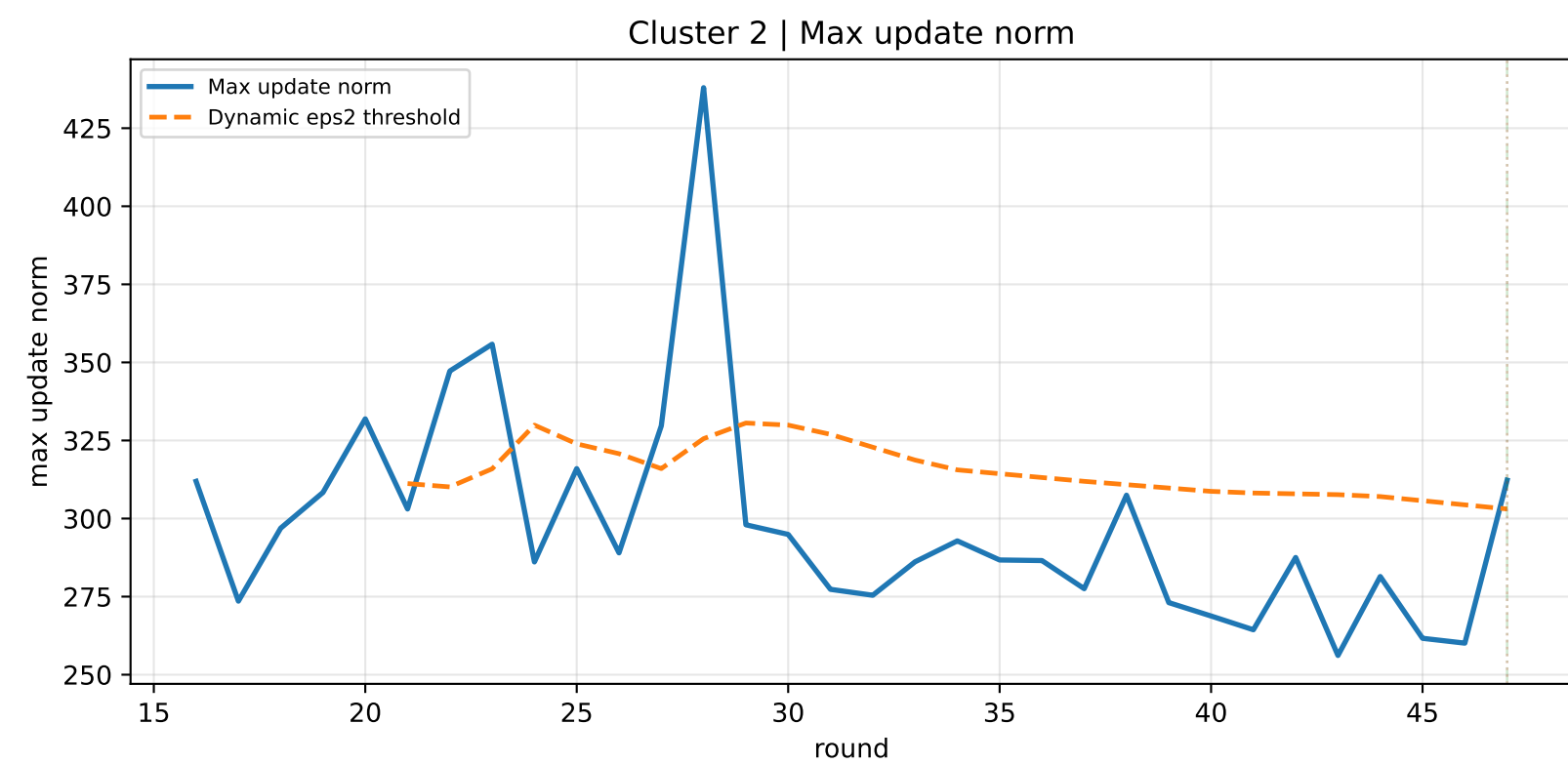
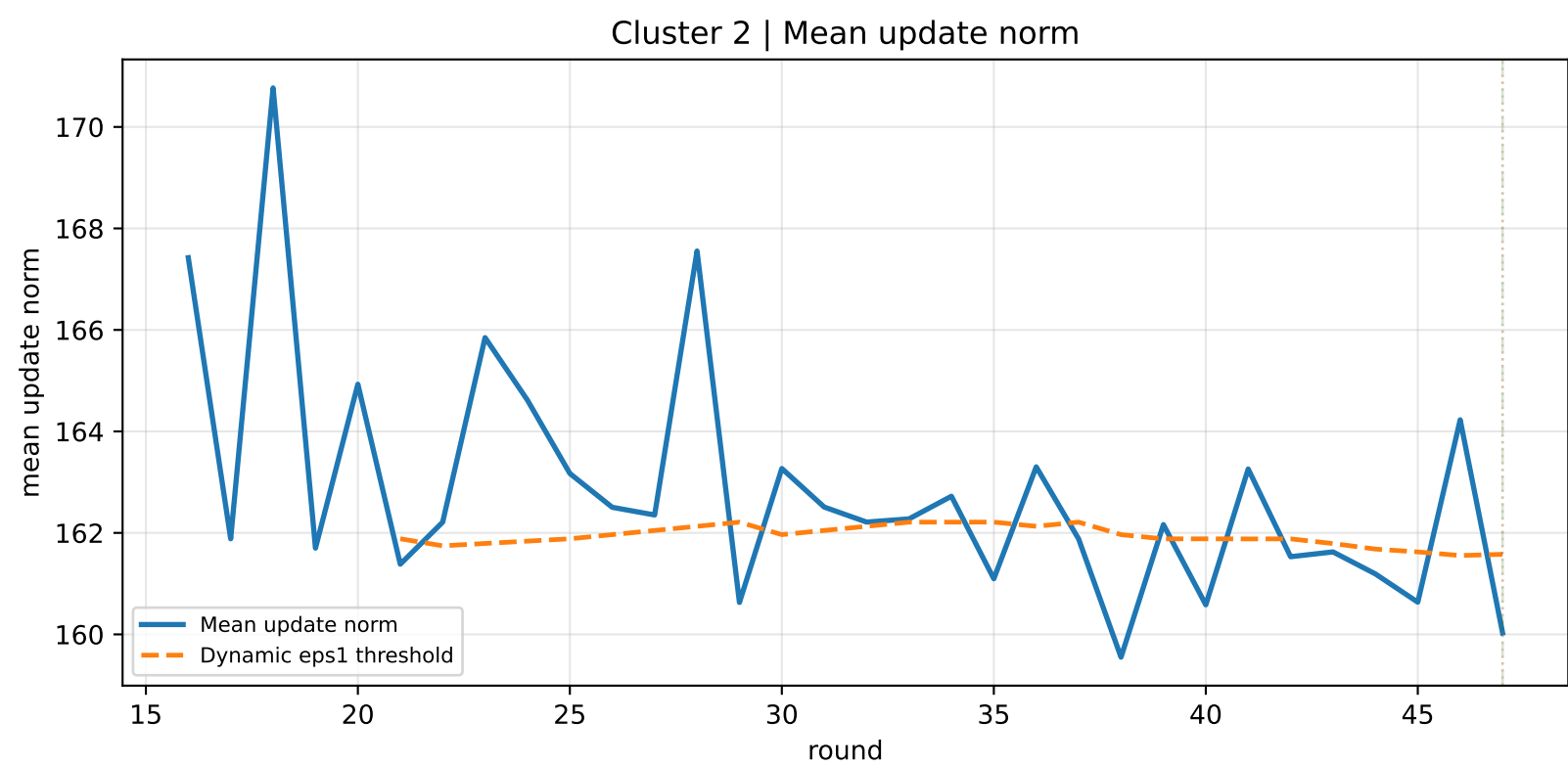
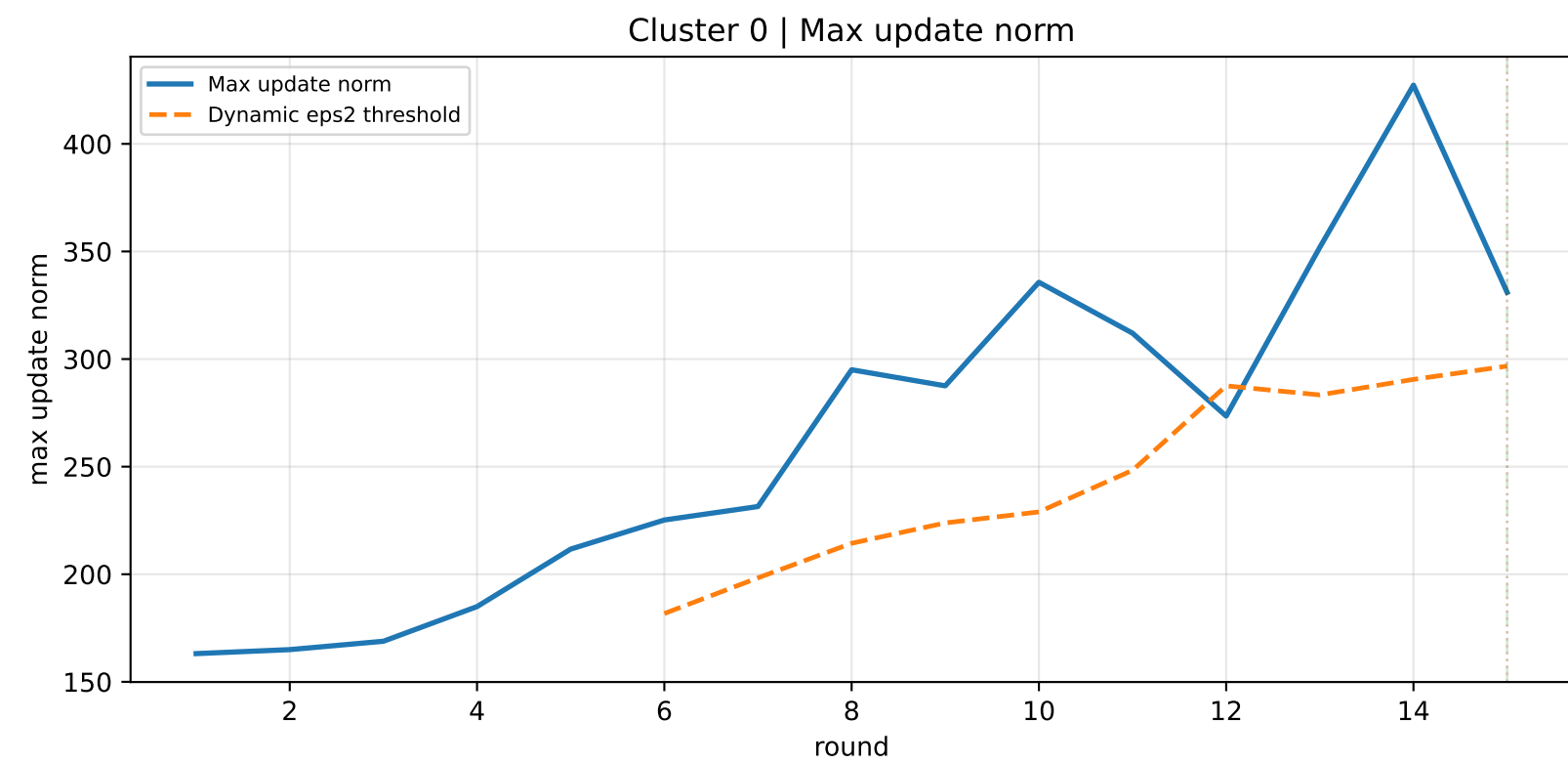
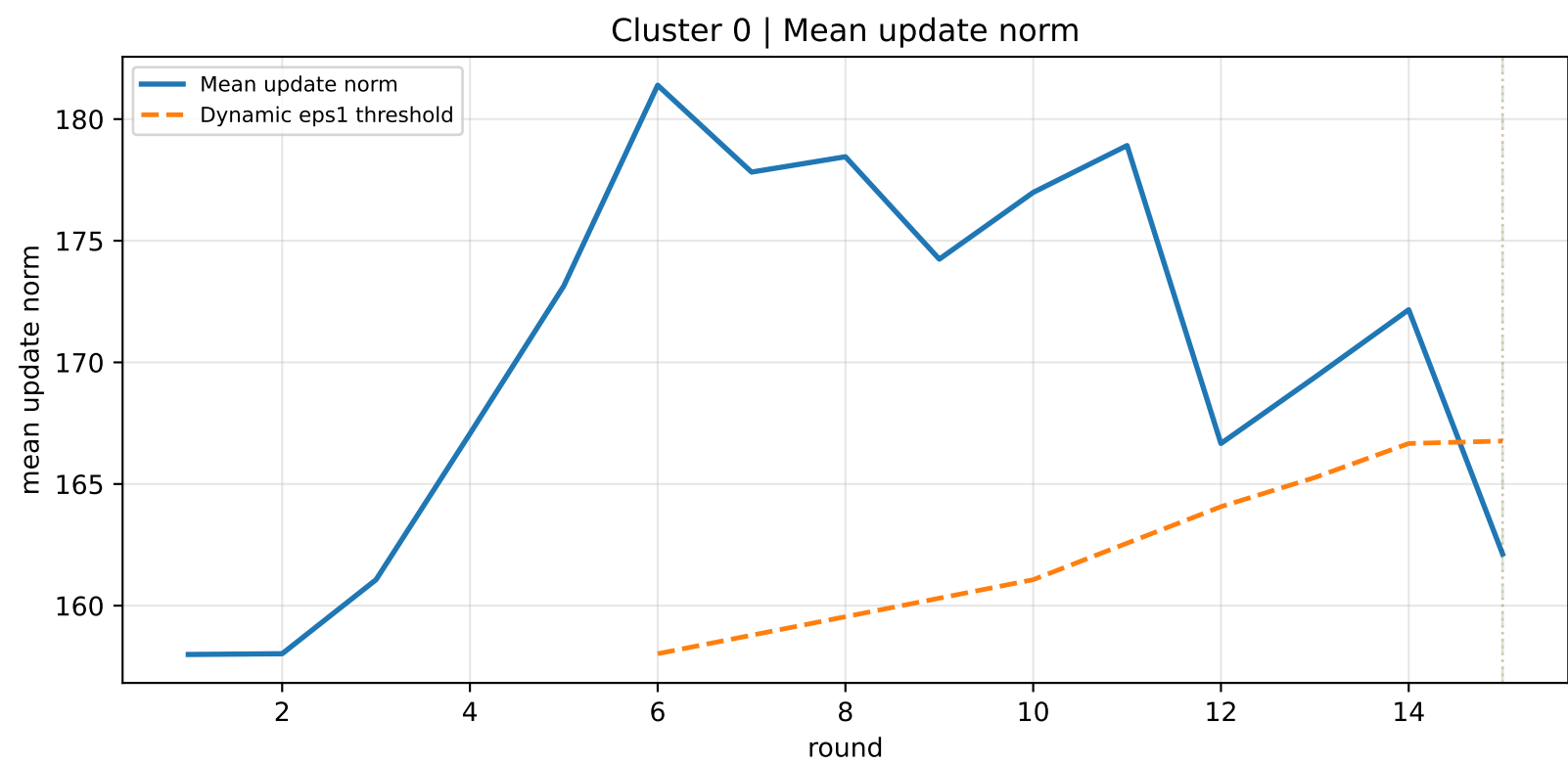
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

Final active clusters | seed=1

cluster_id	cluster_size	members	member_indices	family_mix
1	1	107	18	strong_increasing:1
3	1	0	19	strong_increasing:1
5	1	96	16	strong_increasing:1
6	17	87, 157, 149, 109, 143, 114, 117, 2, 88, 145, 166, 141, 65, 6, 131, 119, 162	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 17	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:1

GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | norm / threshold panels (0-2)

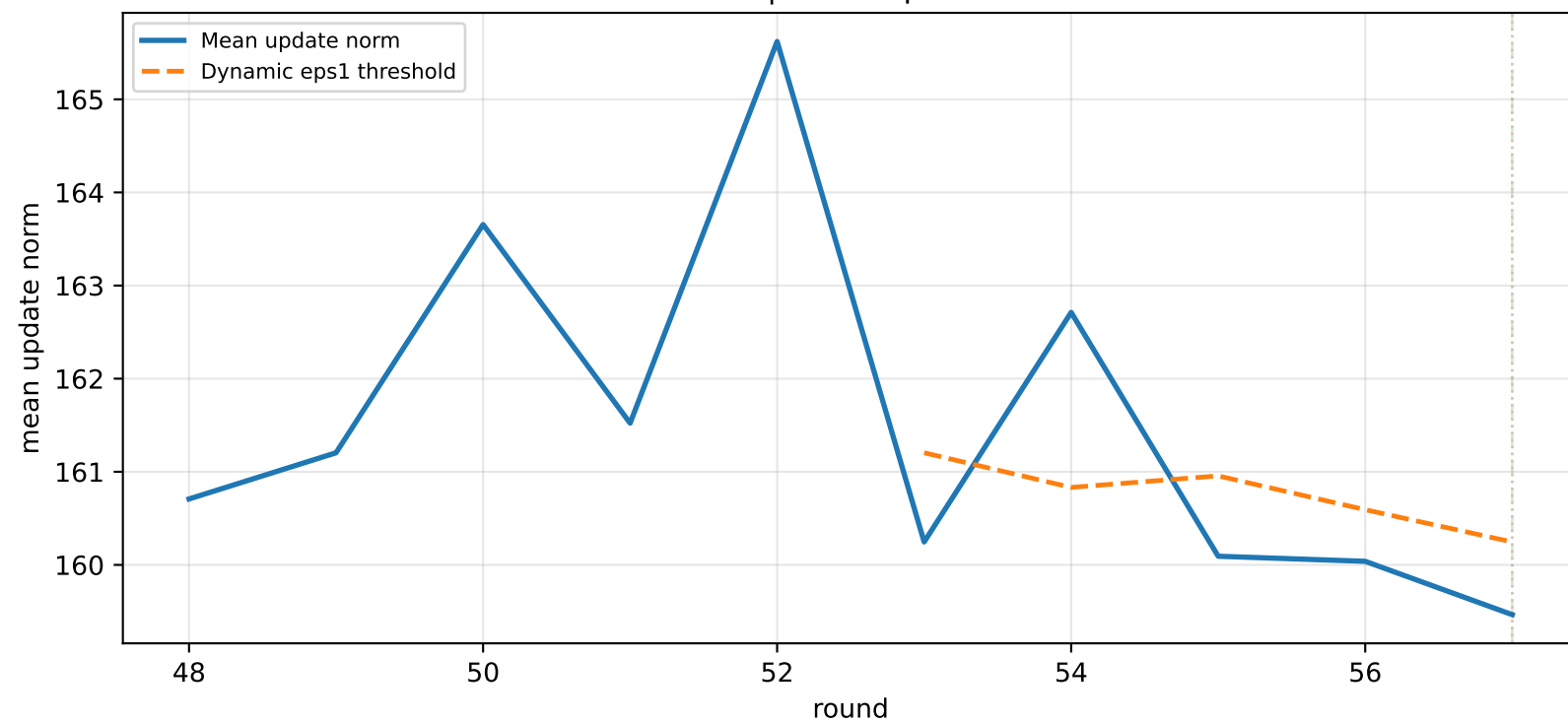
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095



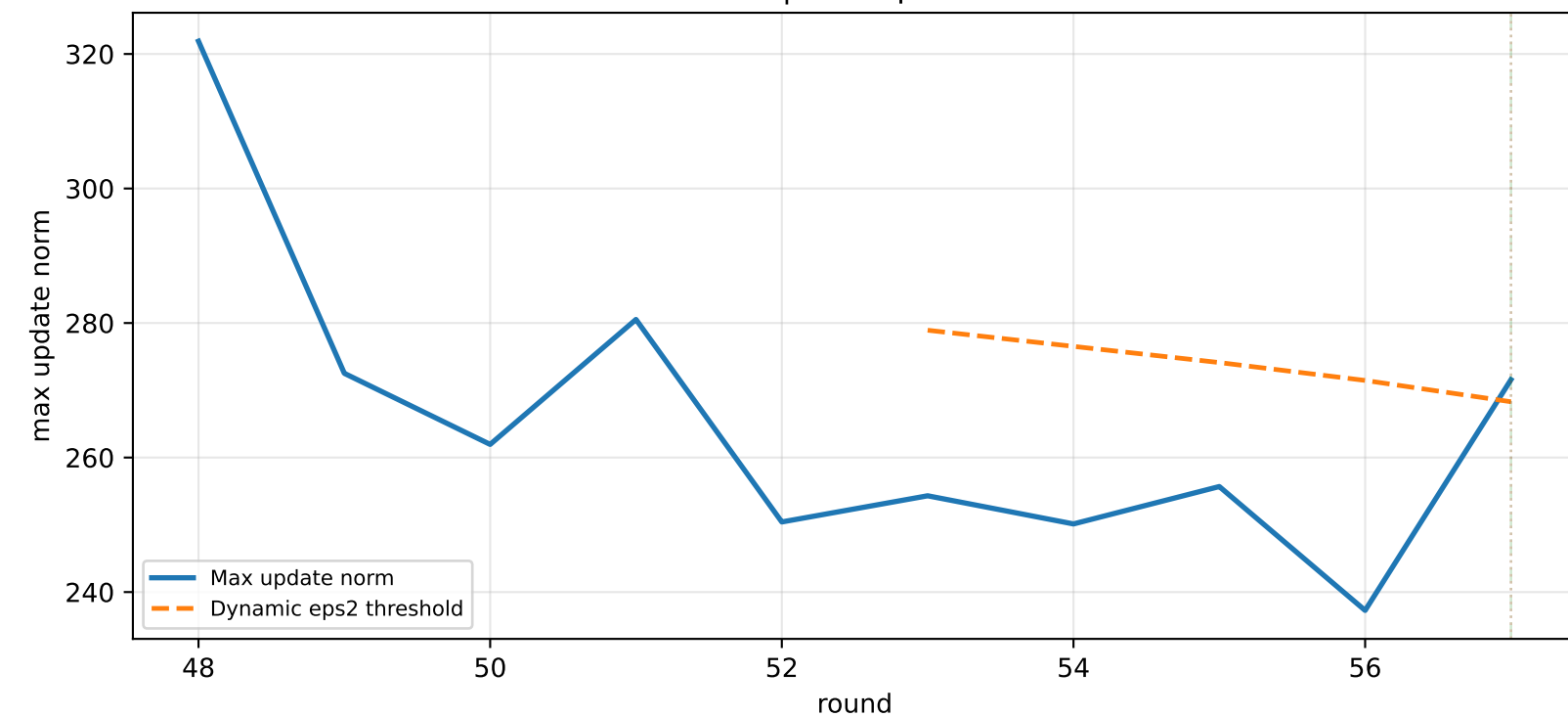
GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=1 | norm / threshold panels (4-6)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

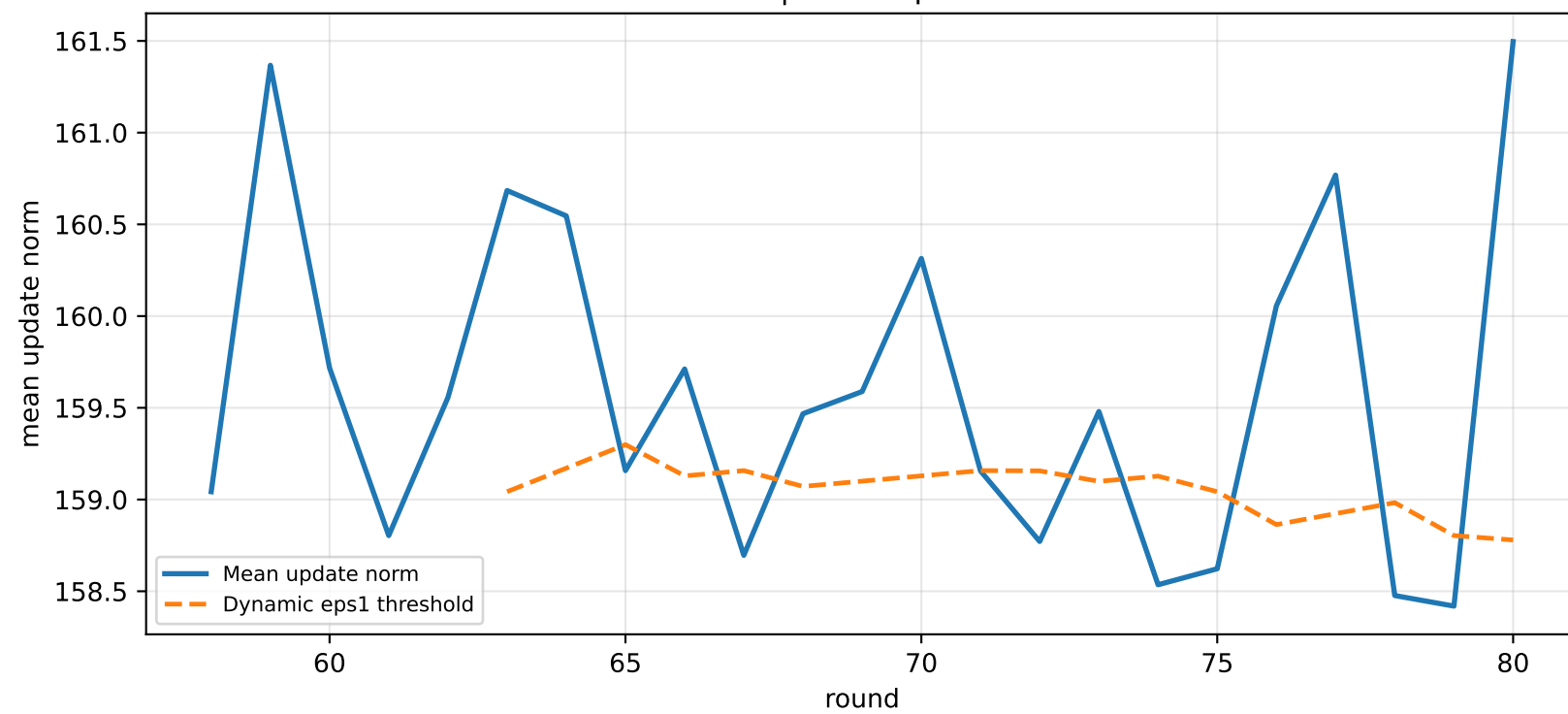
Cluster 4 | Mean update norm



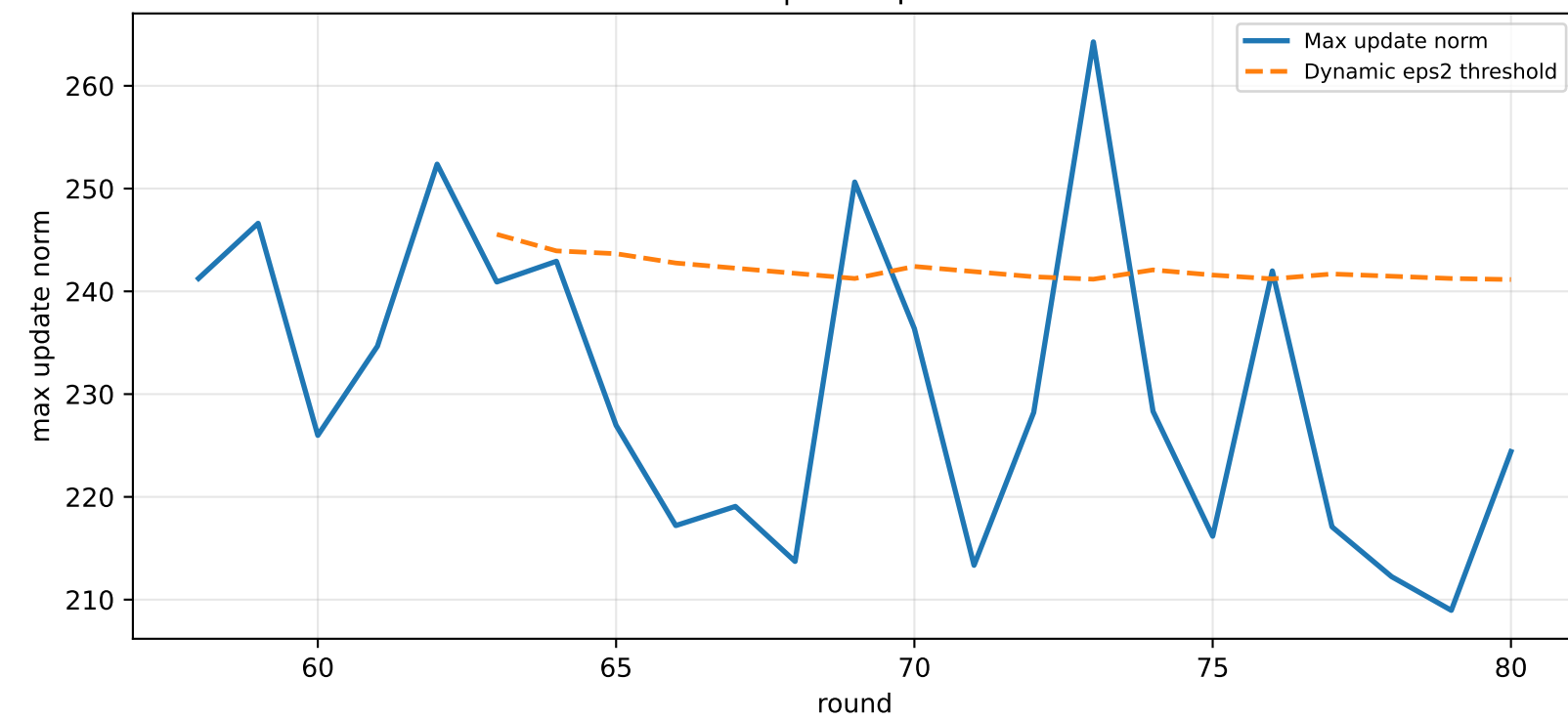
Cluster 4 | Max update norm



Cluster 6 | Mean update norm



Cluster 6 | Max update norm



GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | first applied split diagnostic

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

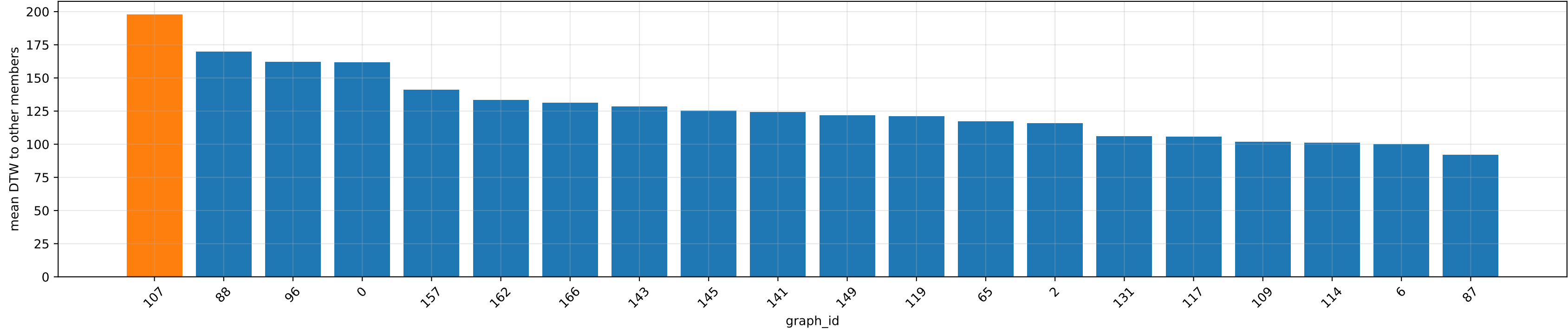
First applied split summary | seed=2

Seed	Round	Parent cluster	Split sizes	Singleton graph	Max mean DTW	Singleton = maxDTW
2	23	0	1 vs 19	107	107	yes

DTW separation diagnostics

Cross DTW	Left DTW	Right DTW
197.861	nan	120.176

Per-member mean DTW



GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | first applied split pairwise DTW table

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

First applied split pairwise DTW metadata | seed=2

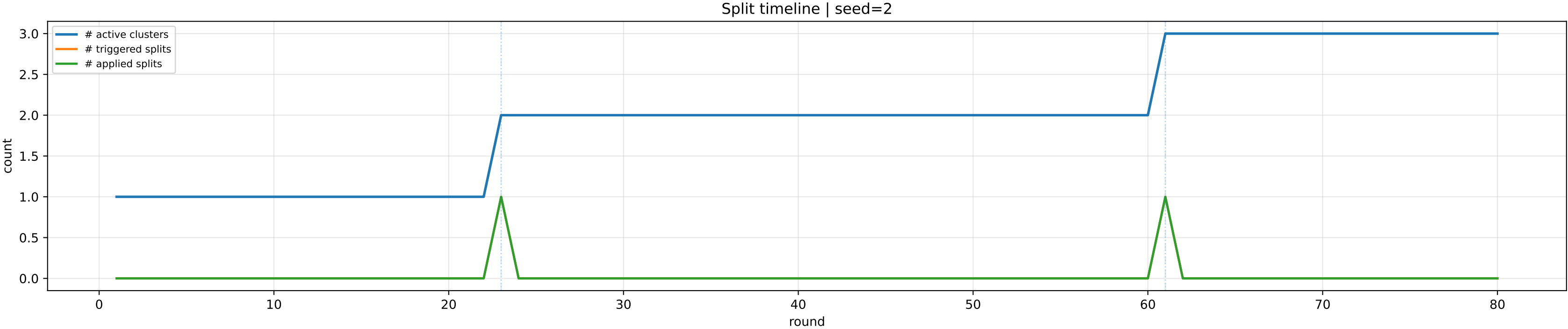
Seed	Round	Parent cluster	n_members	Singleton	Cut value	Cross DTW	Pairwise logged
2	23	0	20	107.0	0.100	197.861	1

Pairwise DTW table | first applied split | seed=2
Global max DTW = 277.9 at (65, 107)

graph_id	87	157	149	109	143	114	117	2	88	145	166	141	65	6	131	119	96	162	107	0
87	0.0	94.3	105.0	61.7	105.5	53.1	65.3	81.1	129.7	75.2	96.3	70.3	66.4	33.1	70.8	87.1	137.6	90.9	183.7	144.1
157	94.3	0.0	90.5	81.1	173.8	153.7	94.9	176.9	245.4	179.4	231.1	138.7	206.2	147.8	66.0	189.6	98.5	72.3	145.8	97.7
149	105.0	90.5	0.0	94.9	147.4	93.7	87.0	157.6	207.3	162.9	165.7	122.3	151.3	121.9	64.5	135.3	59.8	83.5	131.1	135.1
109	61.7	81.1	94.9	0.0	104.2	99.5	72.9	108.1	179.4	103.5	139.0	61.3	118.8	84.3	38.0	128.2	113.8	64.9	166.5	112.2
143	105.5	173.8	147.4	104.2	0.0	87.6	115.1	54.1	179.0	124.9	76.4	81.9	59.7	69.3	114.2	109.4	207.3	131.2	274.1	225.6
114	53.1	153.7	93.7	99.5	87.6	0.0	67.7	70.1	123.6	79.3	66.7	101.5	57.4	54.8	102.8	69.1	149.6	132.8	184.4	170.8
117	65.3	94.9	87.0	72.9	115.1	67.7	0.0	109.9	158.0	117.8	129.9	93.4	116.9	82.6	89.9	110.1	96.2	83.3	176.9	141.1
2	81.1	176.9	157.6	108.1	54.1	70.1	109.9	0.0	128.2	74.5	63.3	97.8	36.4	33.8	106.6	79.5	212.8	141.2	268.8	201.8
88	129.7	245.4	207.3	179.4	179.0	123.6	158.0	128.2	0.0	107.4	114.9	174.8	145.6	137.0	191.6	96.6	254.1	256.6	143.2	256.3
145	75.2	179.4	162.9	103.5	124.9	79.3	117.8	74.5	107.4	0.0	56.0	119.4	69.6	76.2	133.6	105.4	214.2	190.5	222.1	168.4
166	96.3	231.1	165.7	139.0	76.4	66.7	129.9	63.3	114.9	56.0	0.0	141.2	53.6	95.0	158.7	84.9	217.5	188.2	243.0	175.1
141	70.3	138.7	122.3	61.3	81.9	101.5	93.4	97.8	174.8	119.4	141.2	0.0	113.1	96.6	79.7	157.4	178.1	133.6	229.2	172.7
65	66.4	206.2	151.3	118.8	59.7	57.4	116.9	36.4	145.6	69.6	53.6	113.1	0.0	22.6	114.9	71.1	211.1	156.4	277.9	180.9
6	33.1	147.8	121.9	84.3	69.3	54.8	82.6	33.8	137.0	76.2	95.0	96.6	22.6	0.0	85.2	60.9	177.1	130.2	227.4	166.5
131	70.8	66.0	64.5	38.0	114.2	102.8	89.9	106.6	191.6	133.6	158.7	79.7	114.9	85.2	0.0	126.6	113.0	81.4	160.7	118.6
119	87.1	189.6	135.3	128.2	109.4	69.1	110.1	79.5	96.6	105.4	84.9	157.4	71.1	60.9	126.6	0.0	183.6	170.5	184.9	149.0
96	137.6	98.5	59.8	113.8	207.3	149.6	96.2	212.8	254.1	214.2	217.5	178.1	211.1	177.1	113.0	183.6	0.0	121.1	168.6	167.2
162	90.9	72.3	83.5	64.9	131.2	132.8	83.3	141.2	256.6	190.5	188.2	133.6	156.4	130.2	81.4	170.5	121.1	0.0	195.1	111.9
107	183.7	145.8	131.1	166.5	274.1	184.4	176.9	268.8	143.2	222.1	243.0	229.2	277.9	227.4	160.7	184.9	168.6	195.1	0.0	176.1
0	144.1	97.7	135.1	112.2	225.6	170.8	141.1	201.8	256.3	168.4	175.1	172.7	180.9	166.5	118.6	149.0	167.2	111.9	176.1	0.0

GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | split
timeline

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

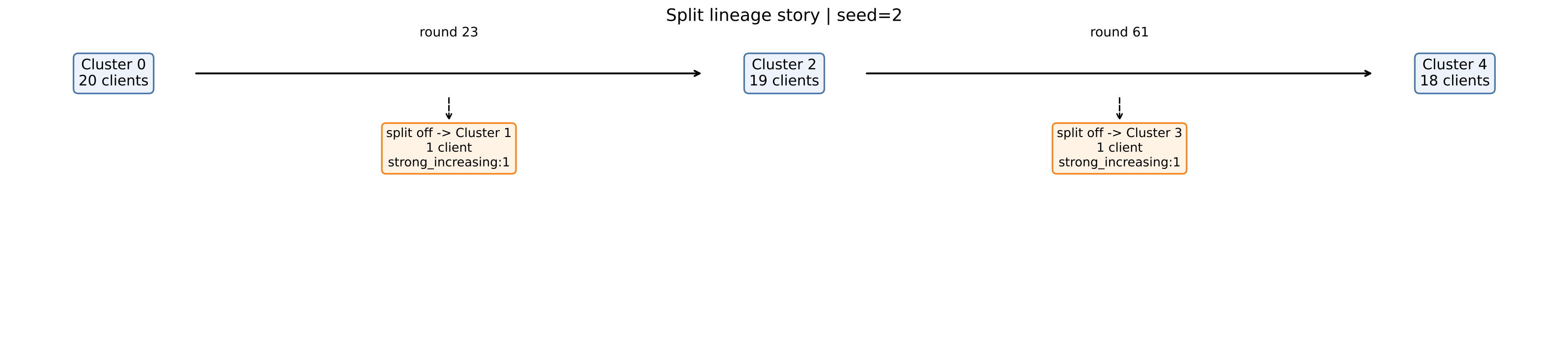


Seed summary | seed=2

Seed	eps1 q	eps2 q	Warmup	Hist	Min cl	Min ch	Triggered	Applied	Final IDs	Final sizes
2	0.25	0.7	5.0	5.0	2.0	1.0	23, 61	23, 61	1, 3, 4	1:1, 3:1, 4:18

GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | split story

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095



Applied split story seed=2							
round	parent cluster	parent size	trigger rule	split-off cluster	split-off family	continuing cluster	continuing family mix
23	0	20	163.242 < 164.443 , 317.849 > 285.333	C1 (1 client)	strong_increasing:1	C2 (19 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:3
61	2	19	160.160 < 160.591 , 293.484 > 288.933	C3 (1 client)	strong_increasing:1	C4 (18 clients)	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:2

GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | final active clusters

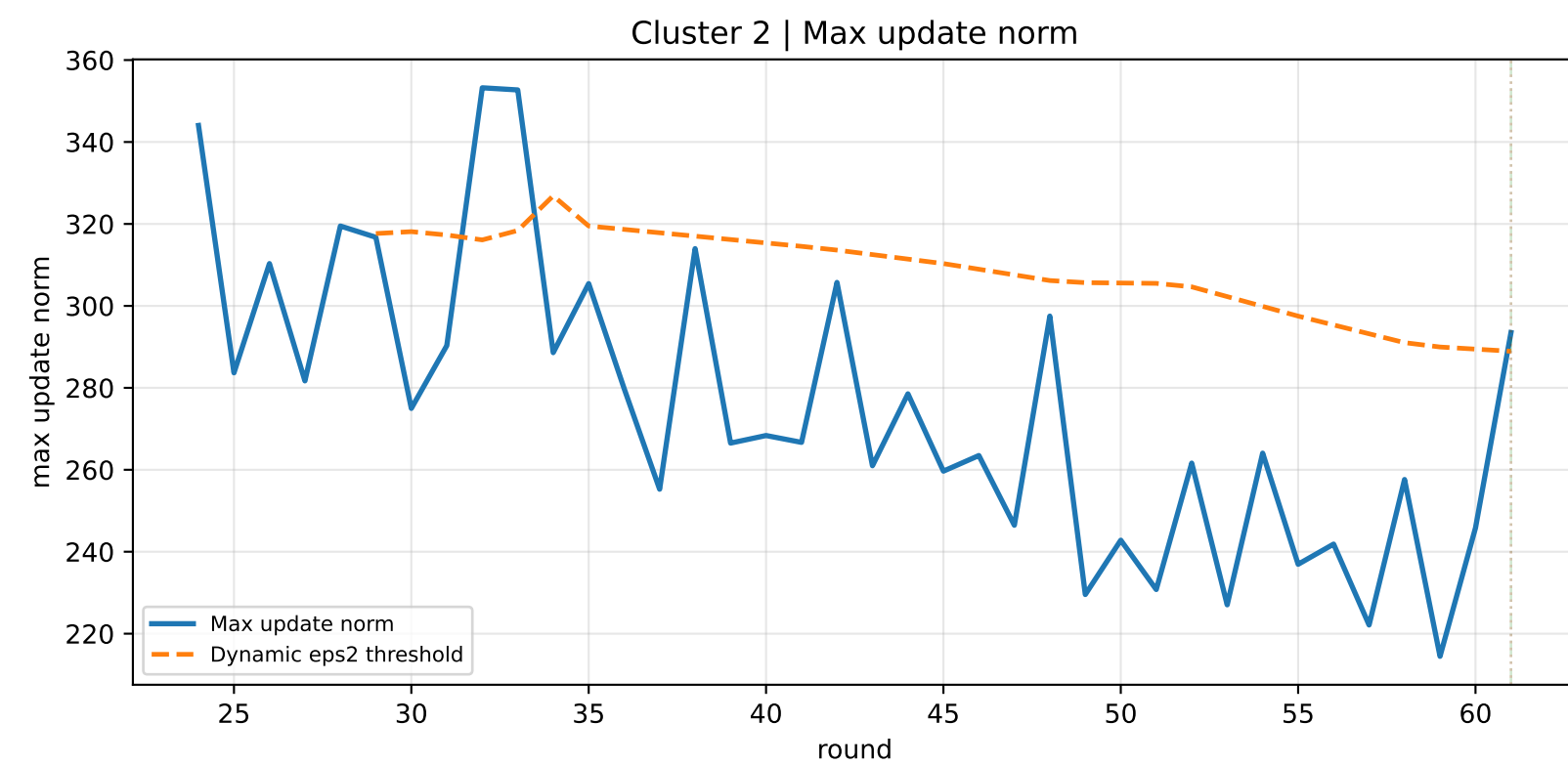
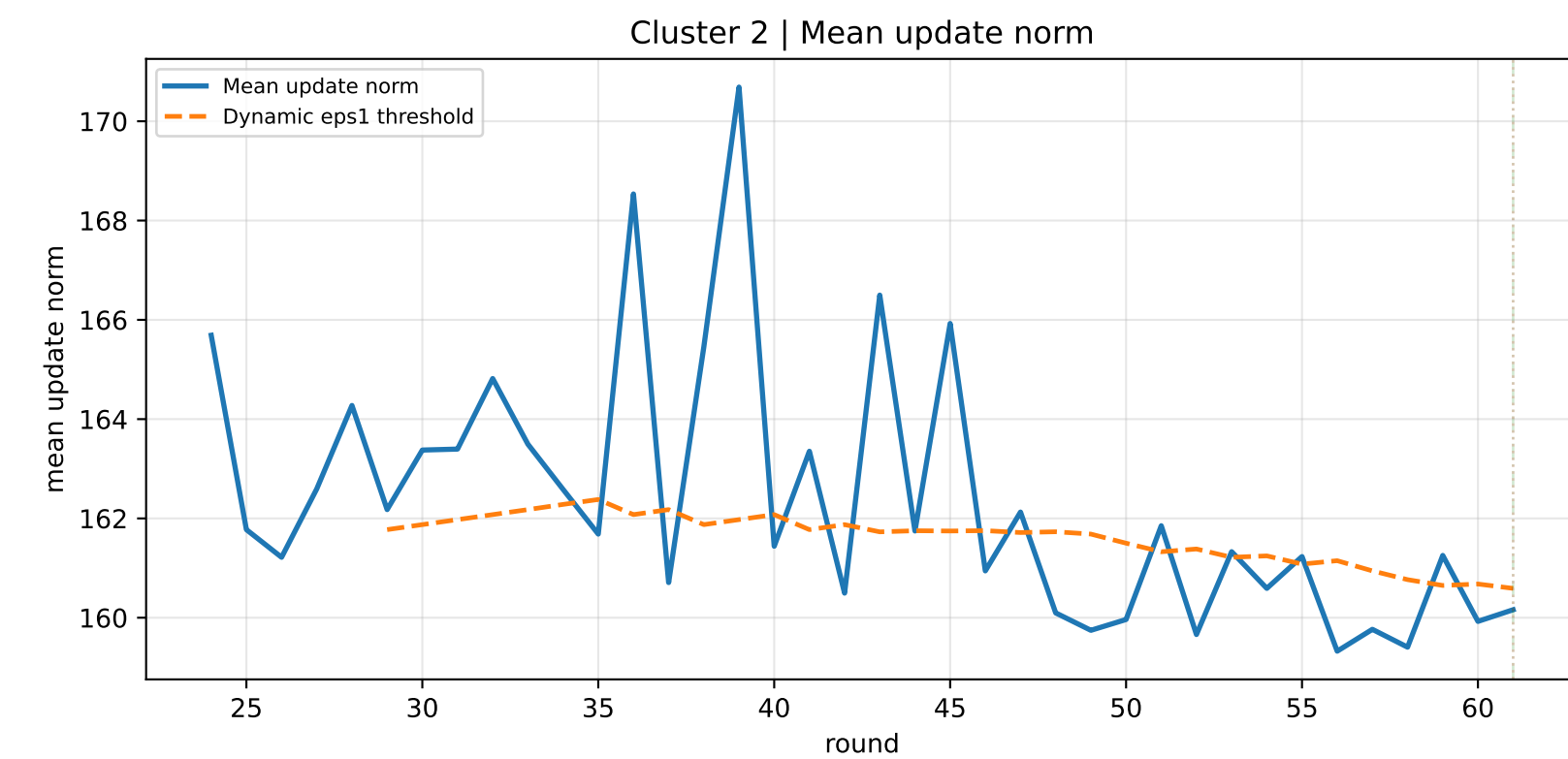
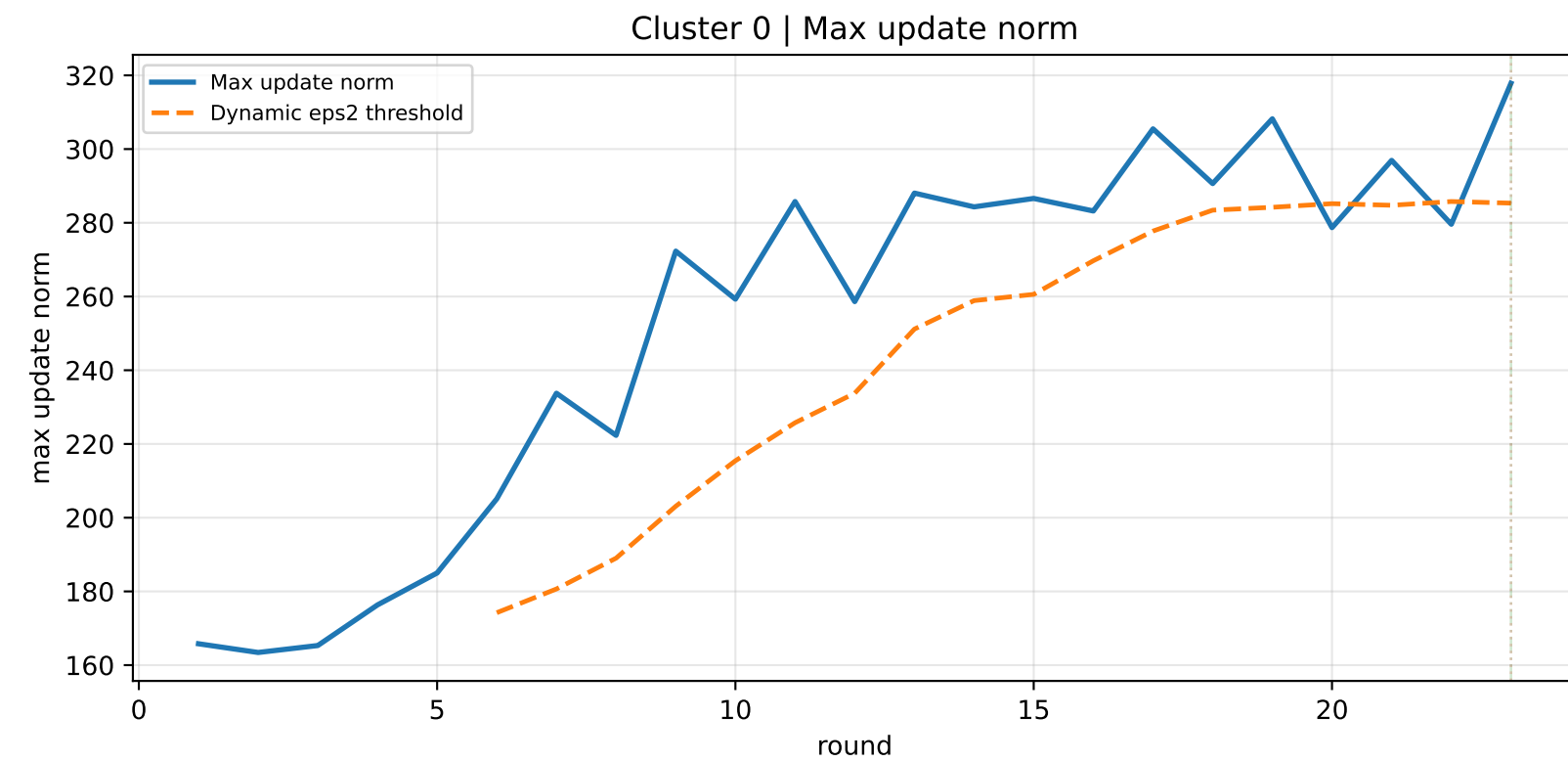
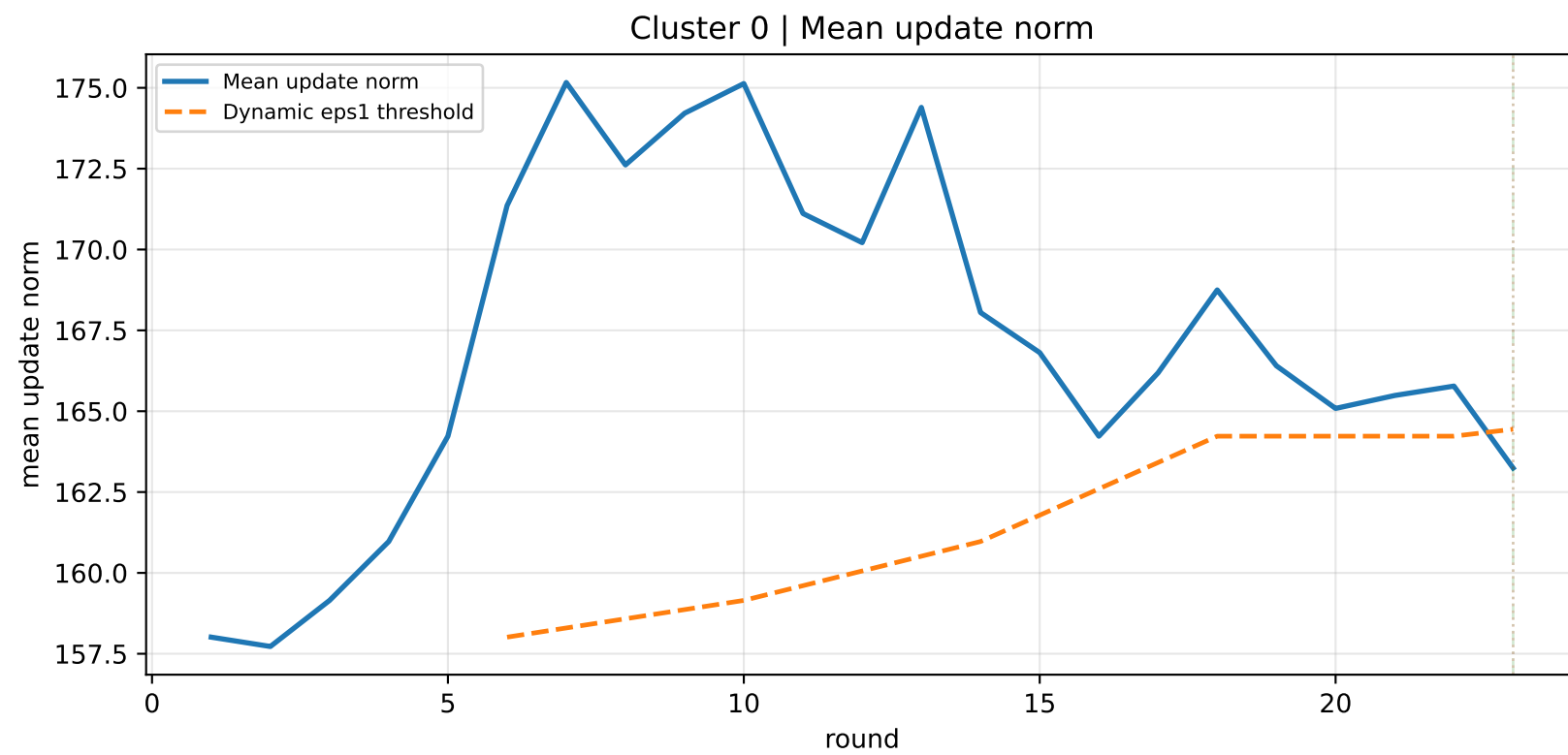
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

Final active clusters | seed=2

cluster_id	cluster_size	members	member_indices	family_mix
1	1	107	18	strong_increasing:1
3	1	162	17	strong_increasing:1
4	18	87, 157, 149, 109, 143, 114, 117, 2, 88, 145, 166, 141, 65, 6, 131, 119, 96, 0	0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 19	flat_balanced:4 , mild_decreasing:4 , mild_increasing:4 , strong_decreasing:4 , strong_increasing:2

GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | norm / threshold panels (0-2)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64
heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

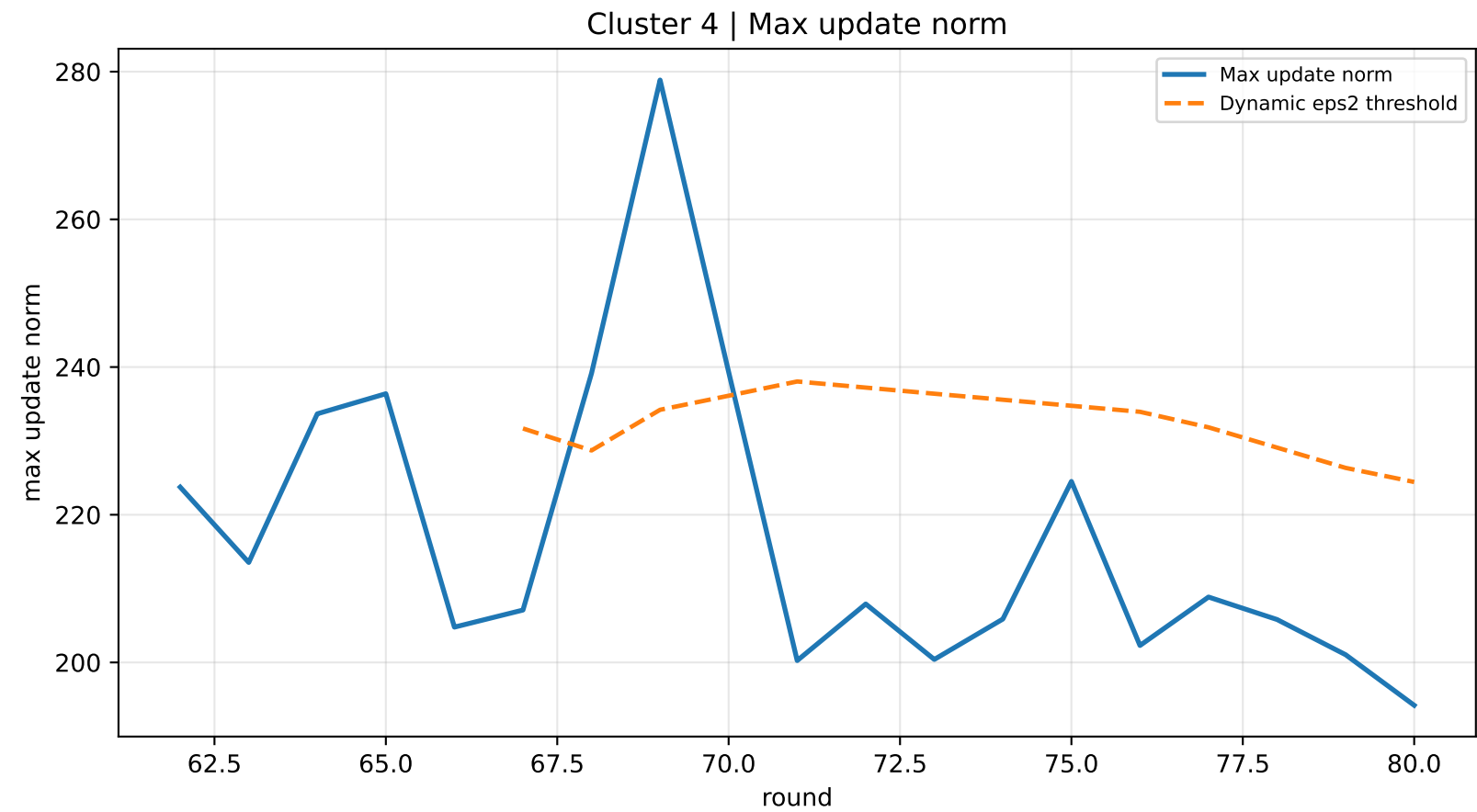
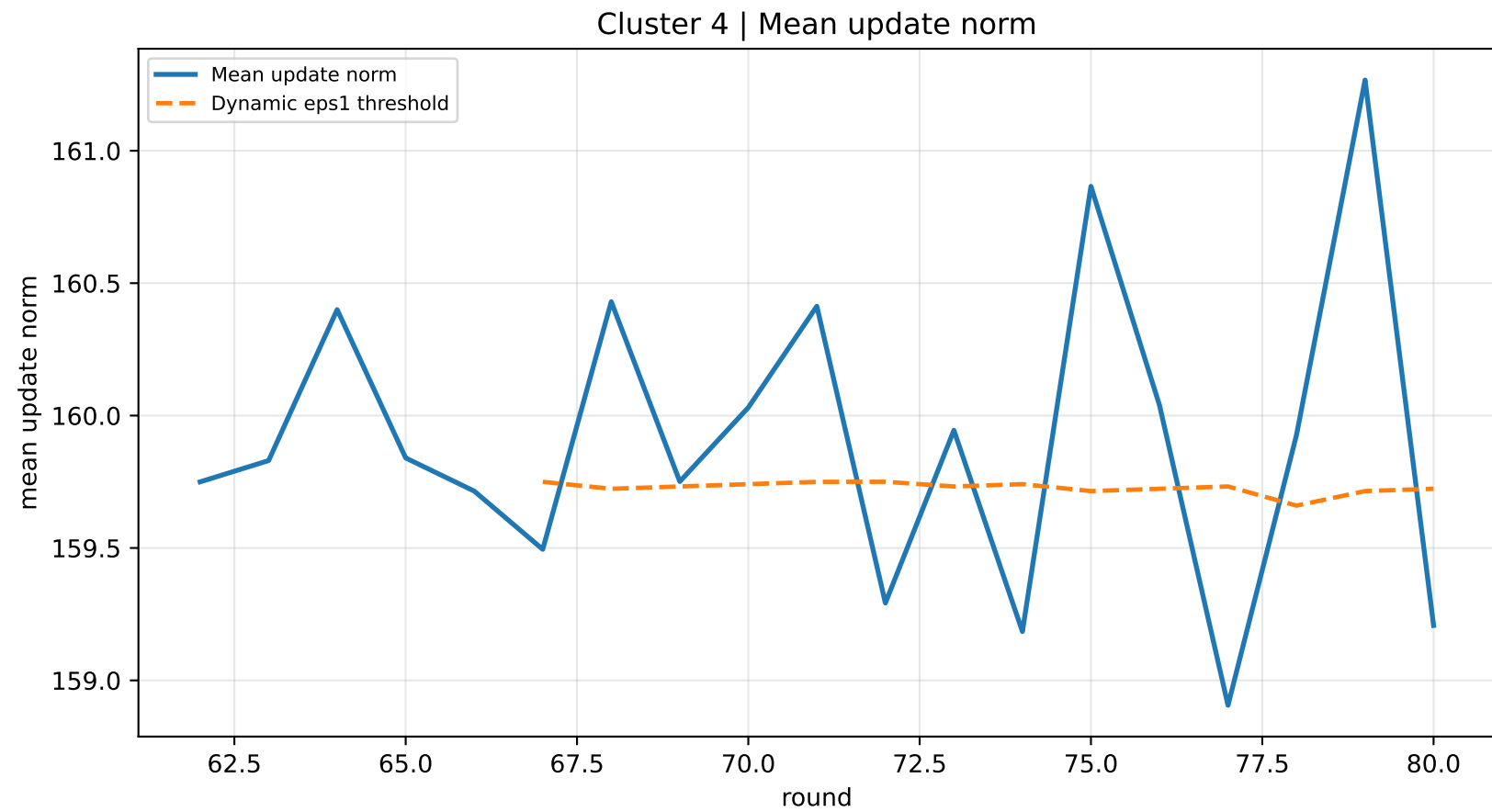


GCFL+ diagnostics | aggregated / pooled over 20 clients | seed=2 | norm / threshold panels (4-4)

model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015

per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095



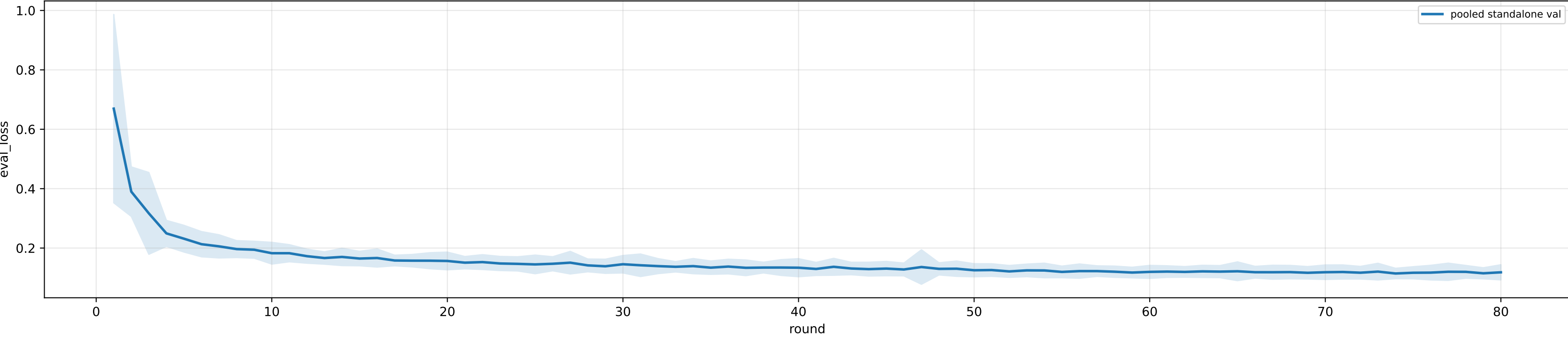
GCFL+ diagnostics | aggregated / pooled over 20 clients |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

heterogeneity | task_jsd_mean=0.013 | task_jsd_median=0.009 | task_jsd_max=0.045 | family_centroid_jsd_mean=0.015
per-task pos-rate std | cycle2_std=0.100 | cycle3_std=0.079 | cycle4_std=0.074 | cycle5_std=0.084 | cycle6_std=0.095

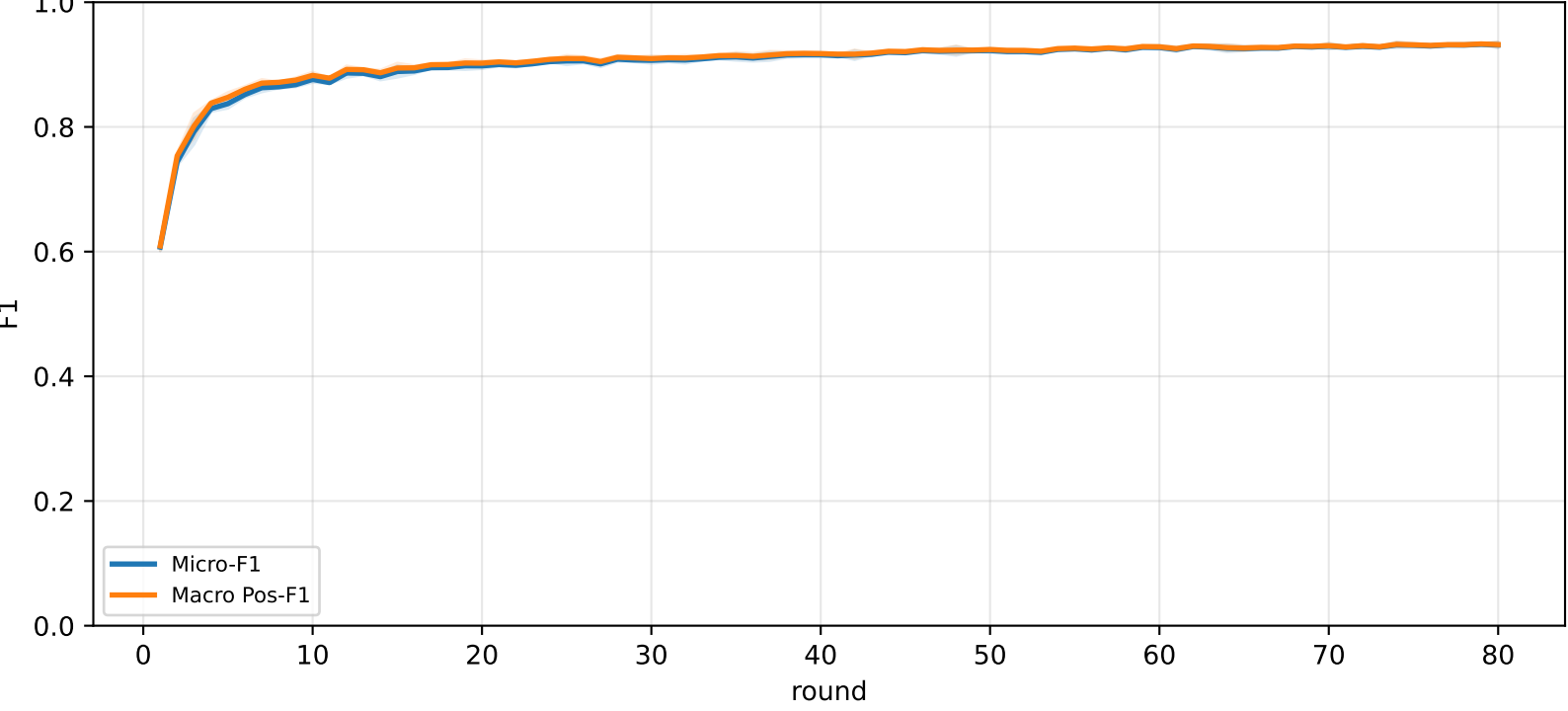
Best logged test performance (GCFL+ vs Local centralized)

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	92.3% ± 0.2%	92.1% ± 0.2%	99.1% ± 0.0%	98.3% ± 0.1%	95.9% ± 0.2%	90.9% ± 0.2%	82.3% ± 0.8%
GCFL+	94.2% ± 0.1%	93.8% ± 0.1%	99.2% ± 0.0%	98.5% ± 0.1%	95.5% ± 0.5%	92.4% ± 0.2%	84.5% ± 0.4%

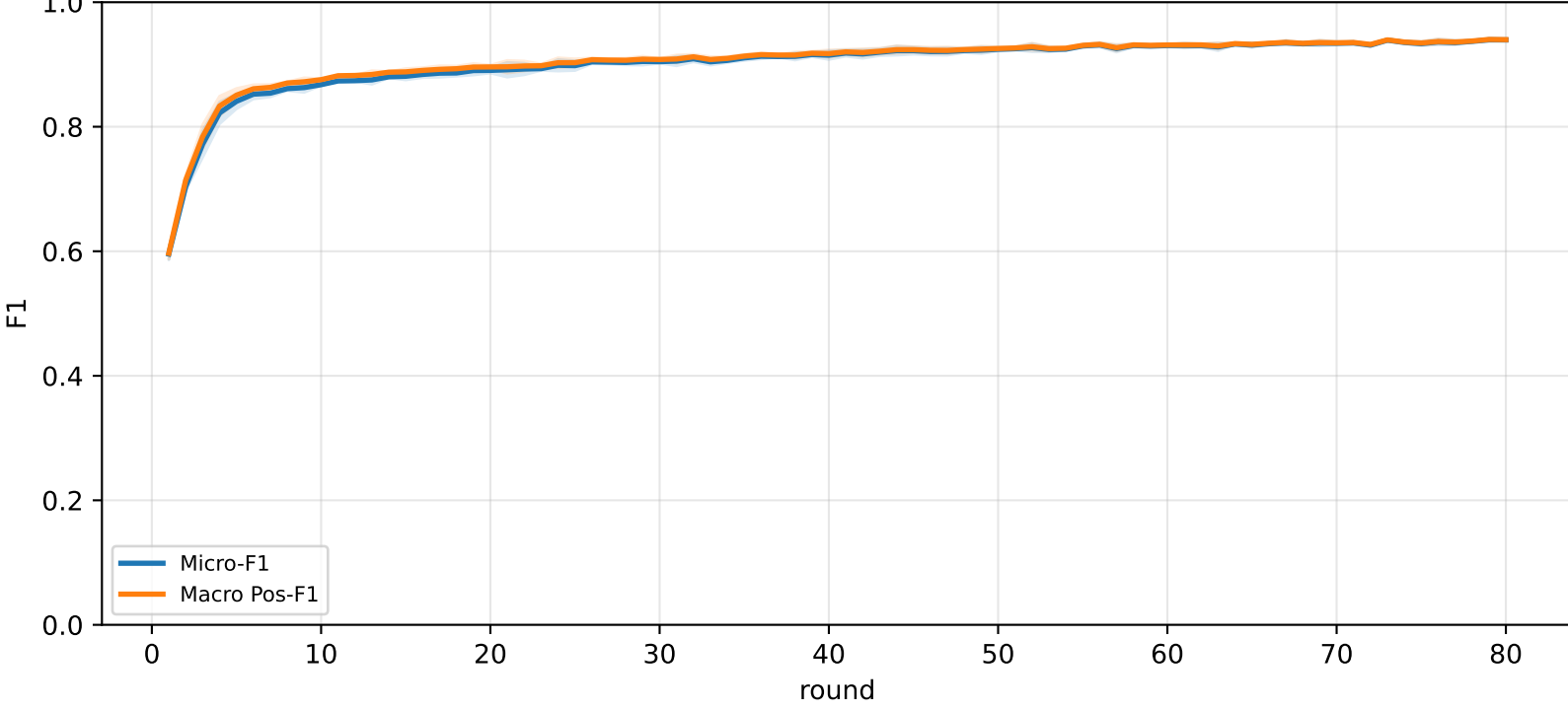
validation loss comparison



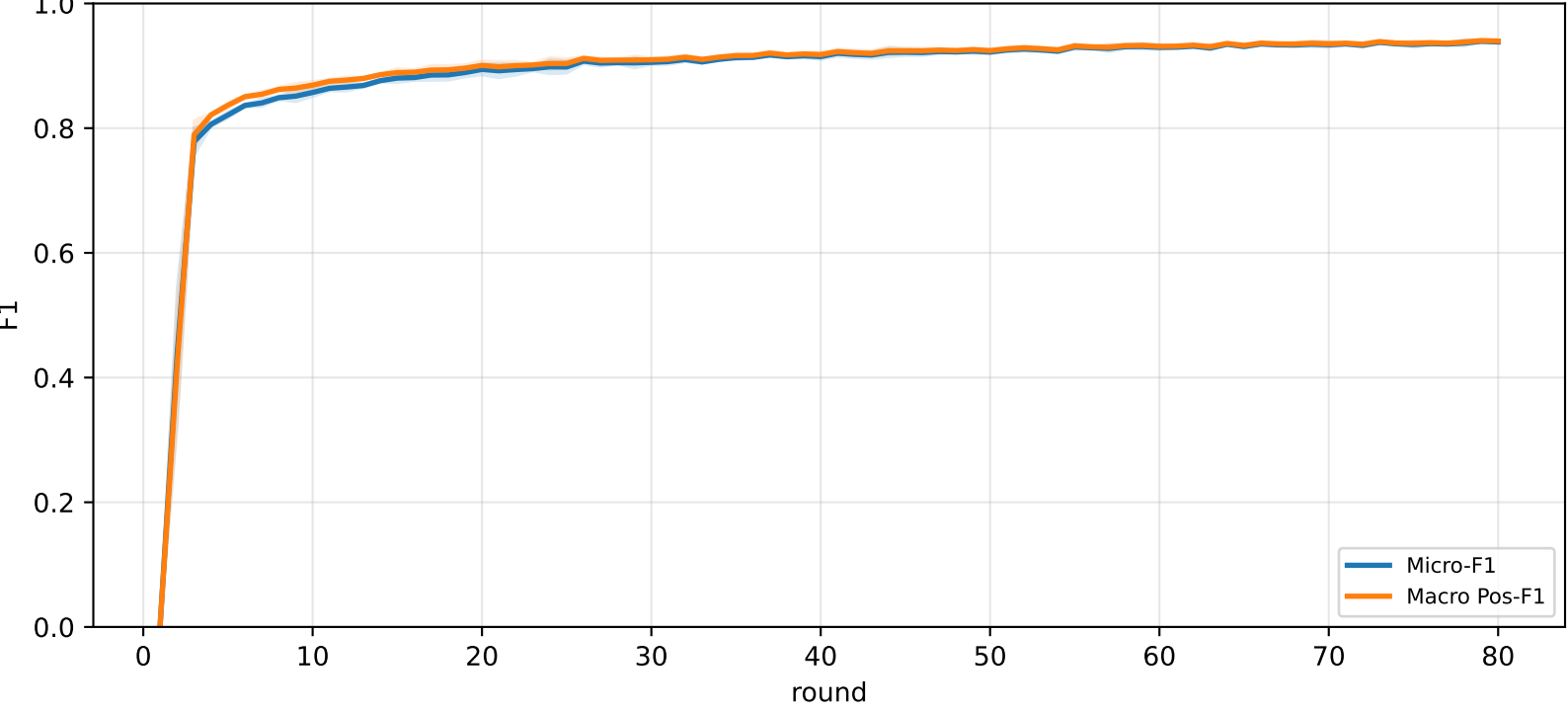
Pooled local centralized micro / macro F1



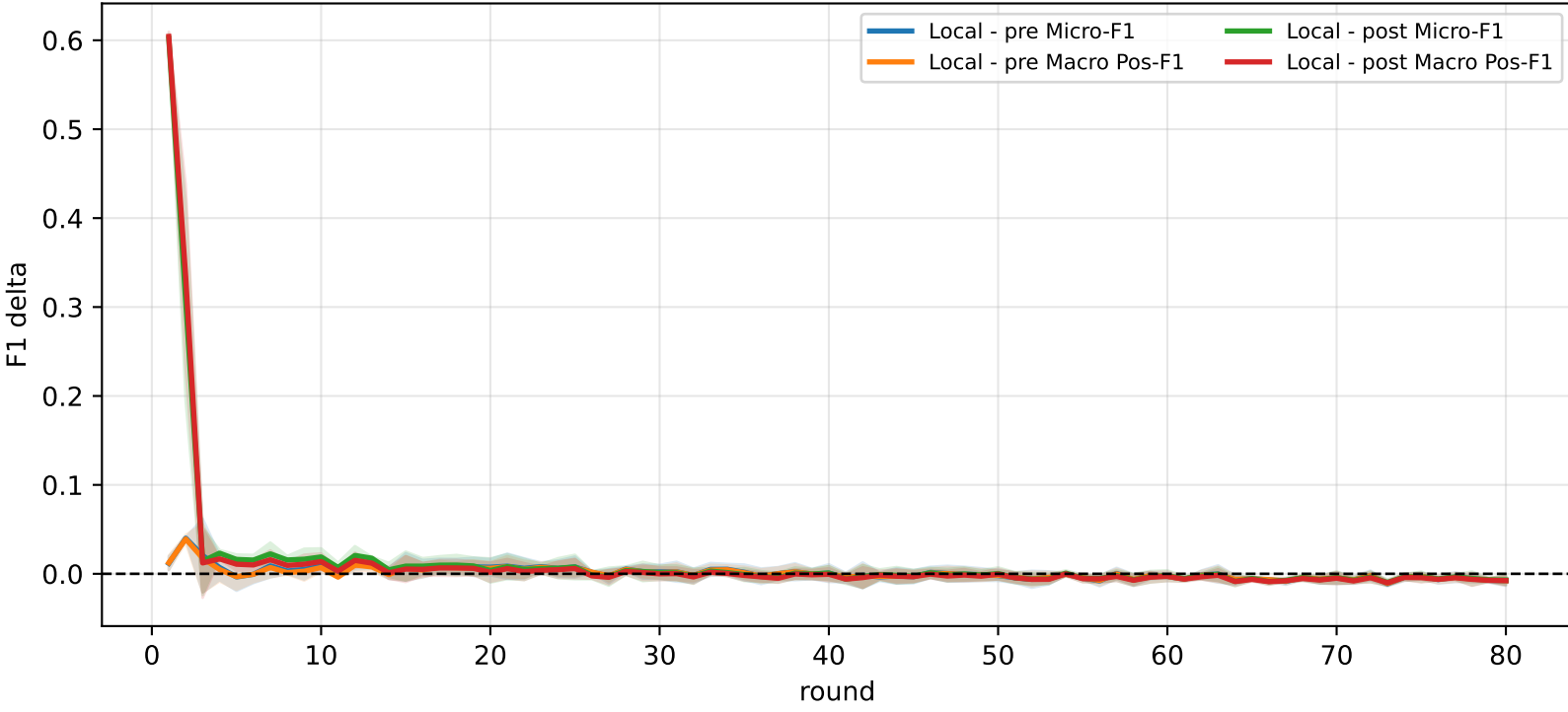
Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



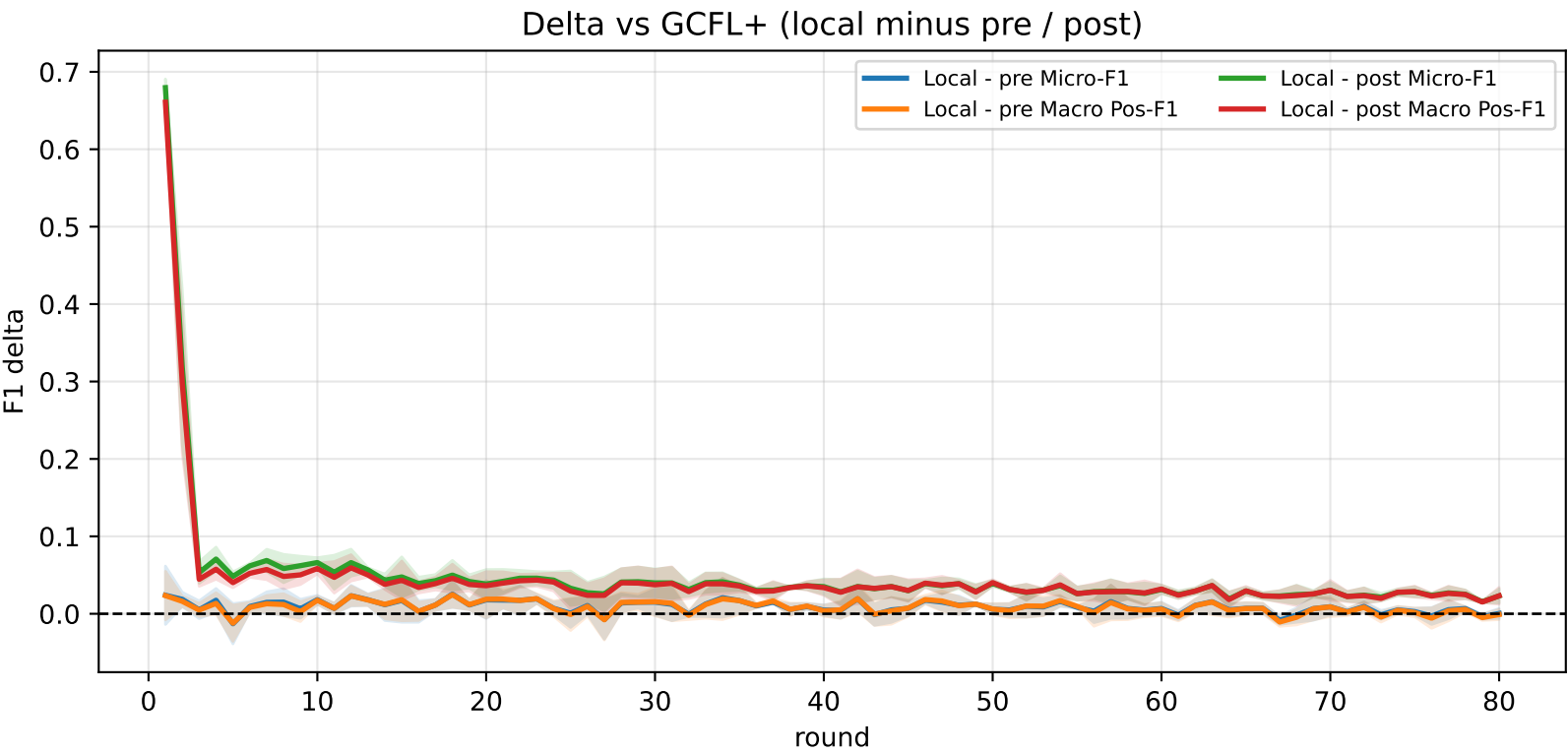
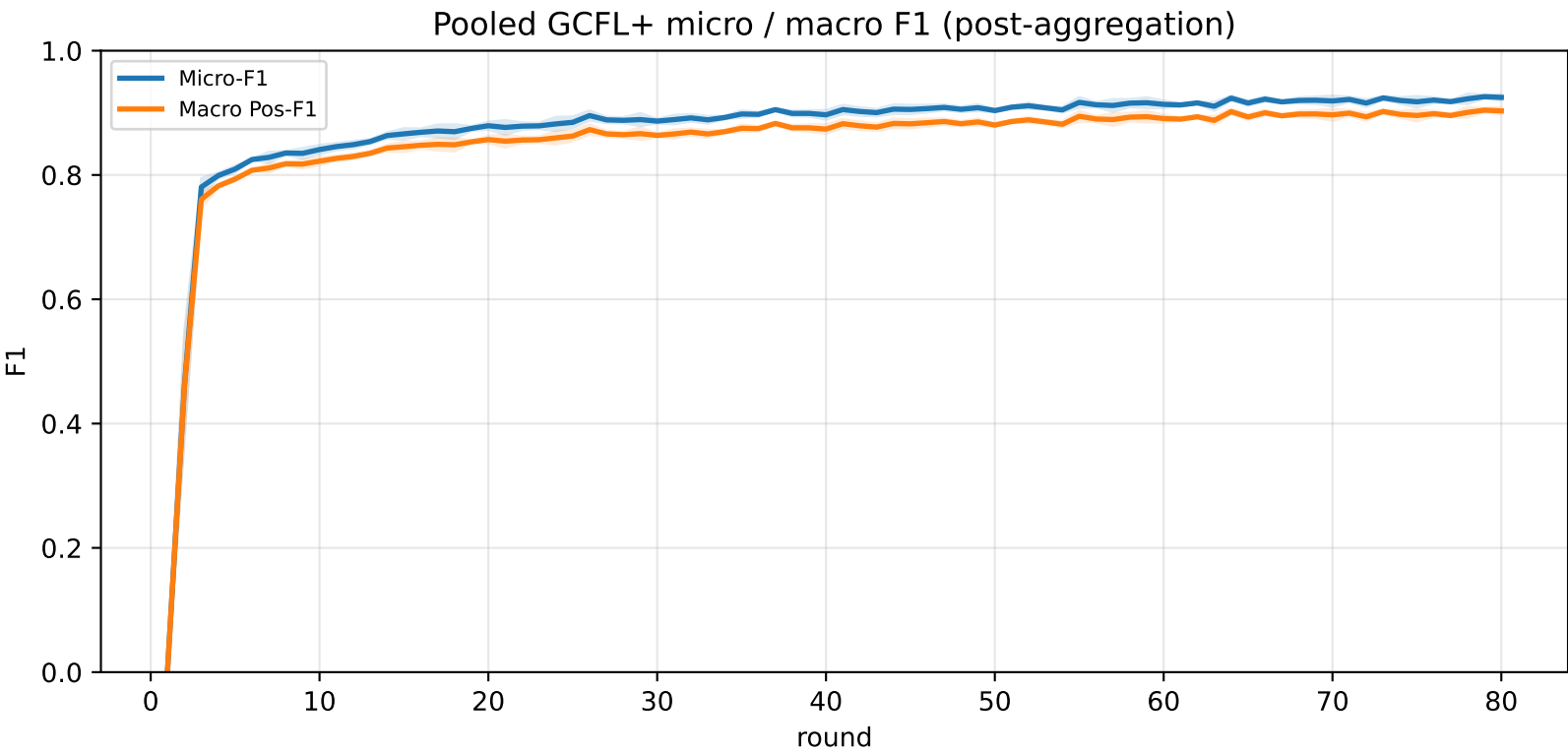
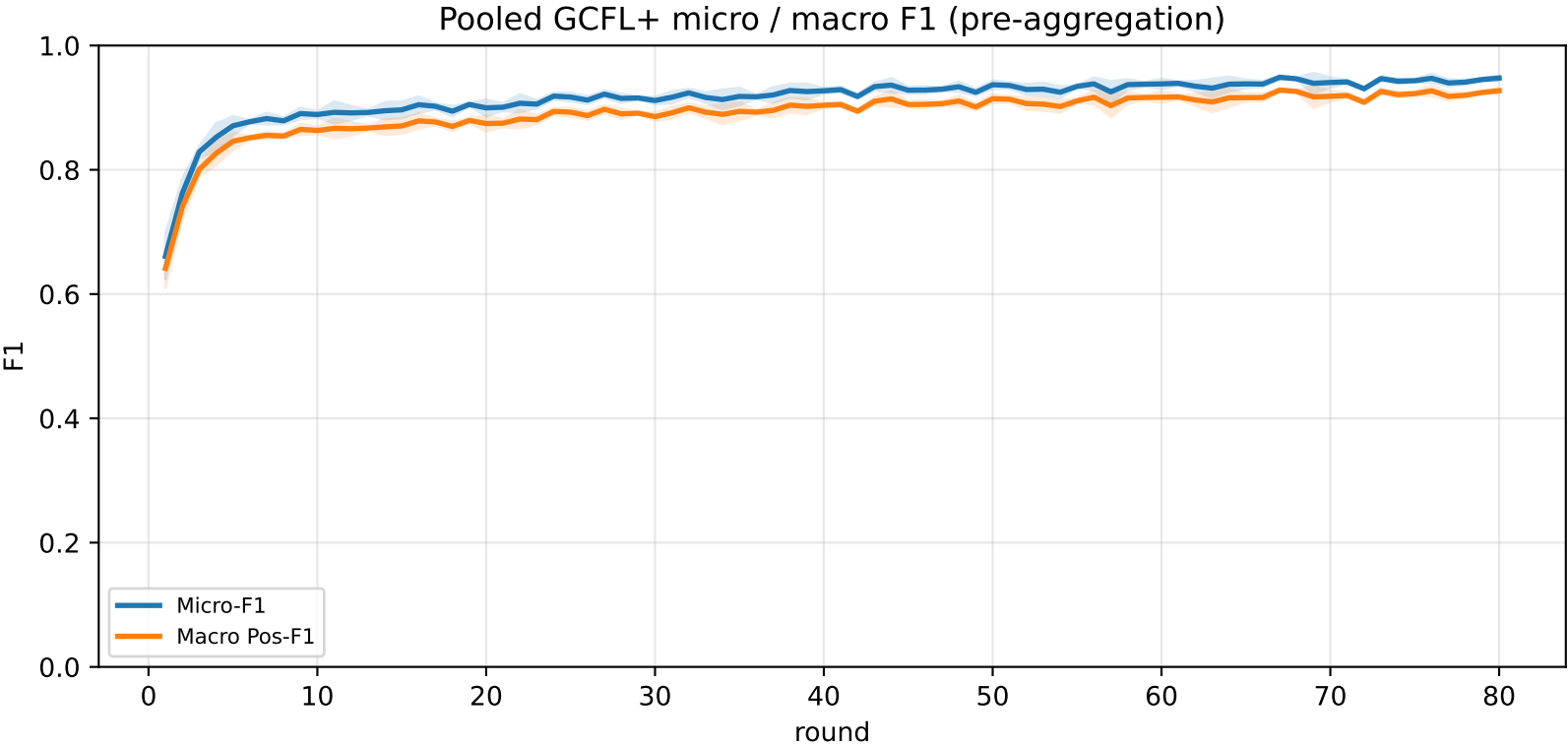
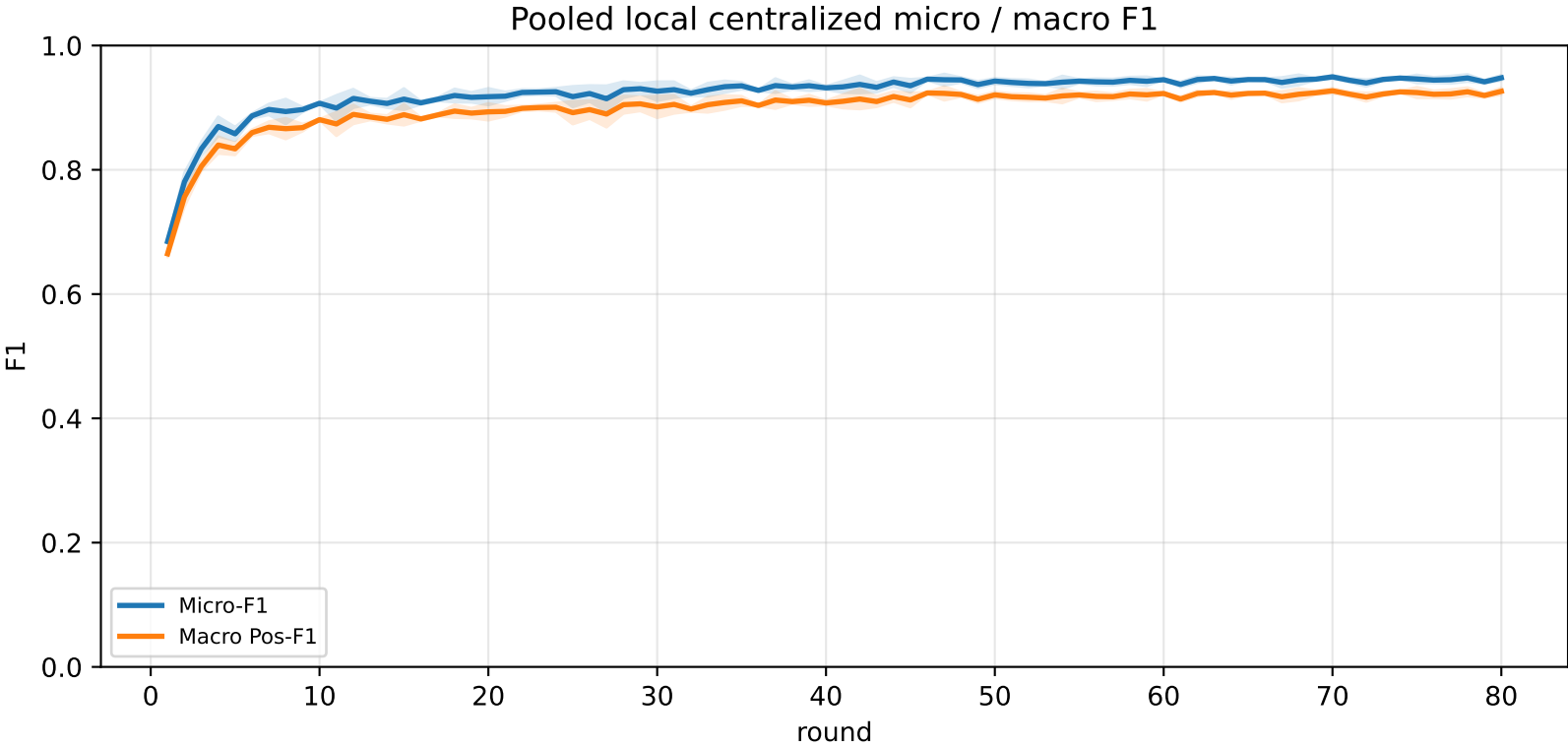
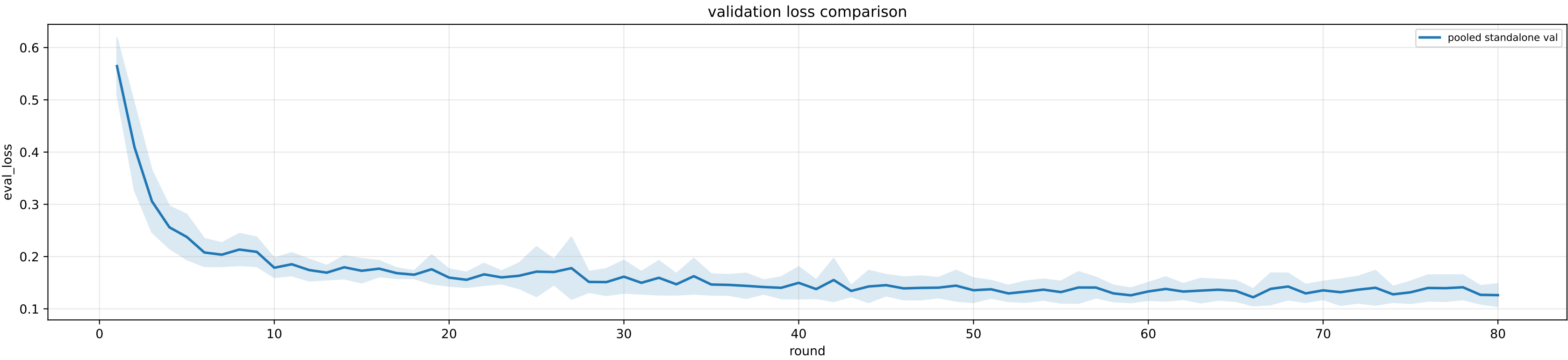
Delta vs GCFL+ (local minus pre / post)



GCFL+ diagnostics | family=strong_decreasing | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

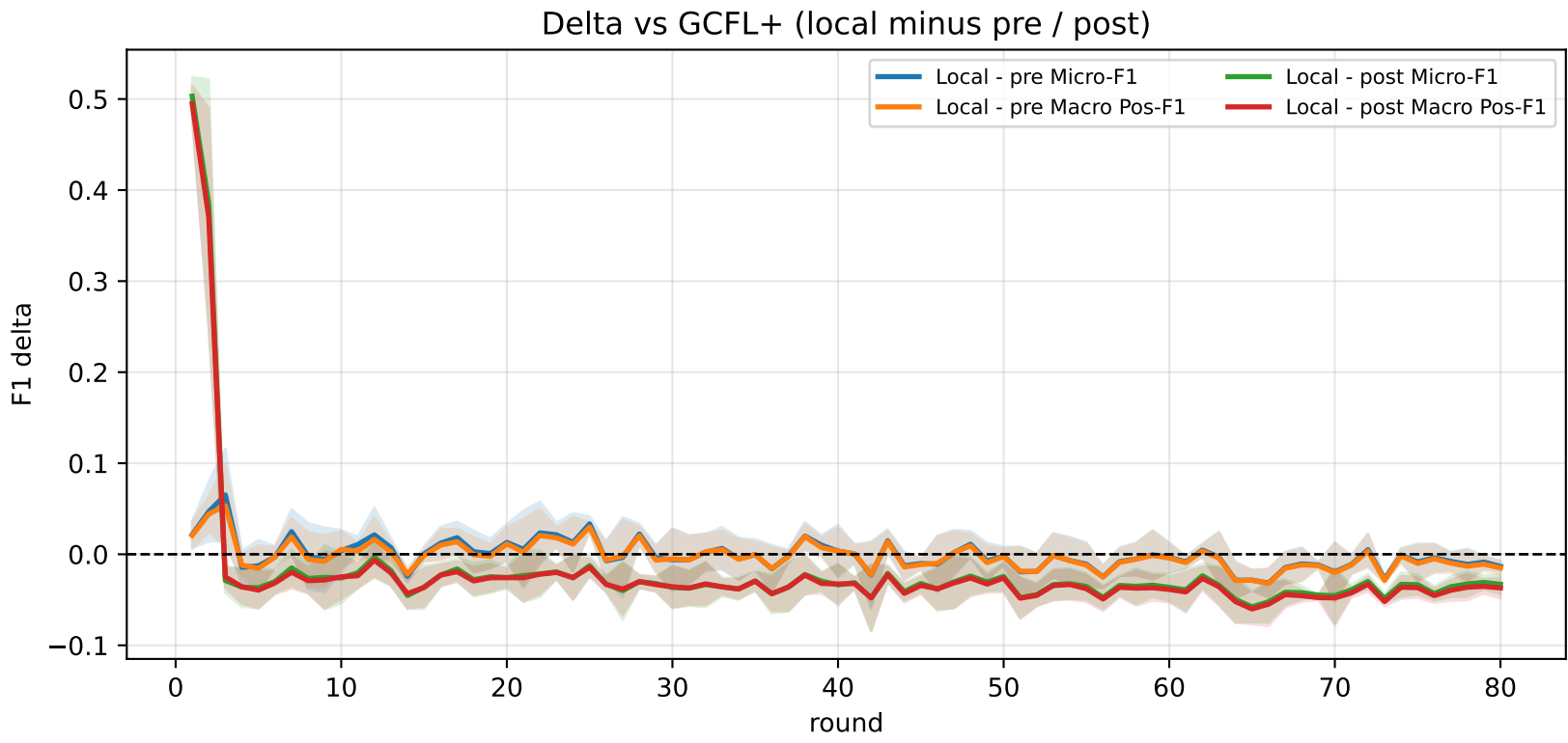
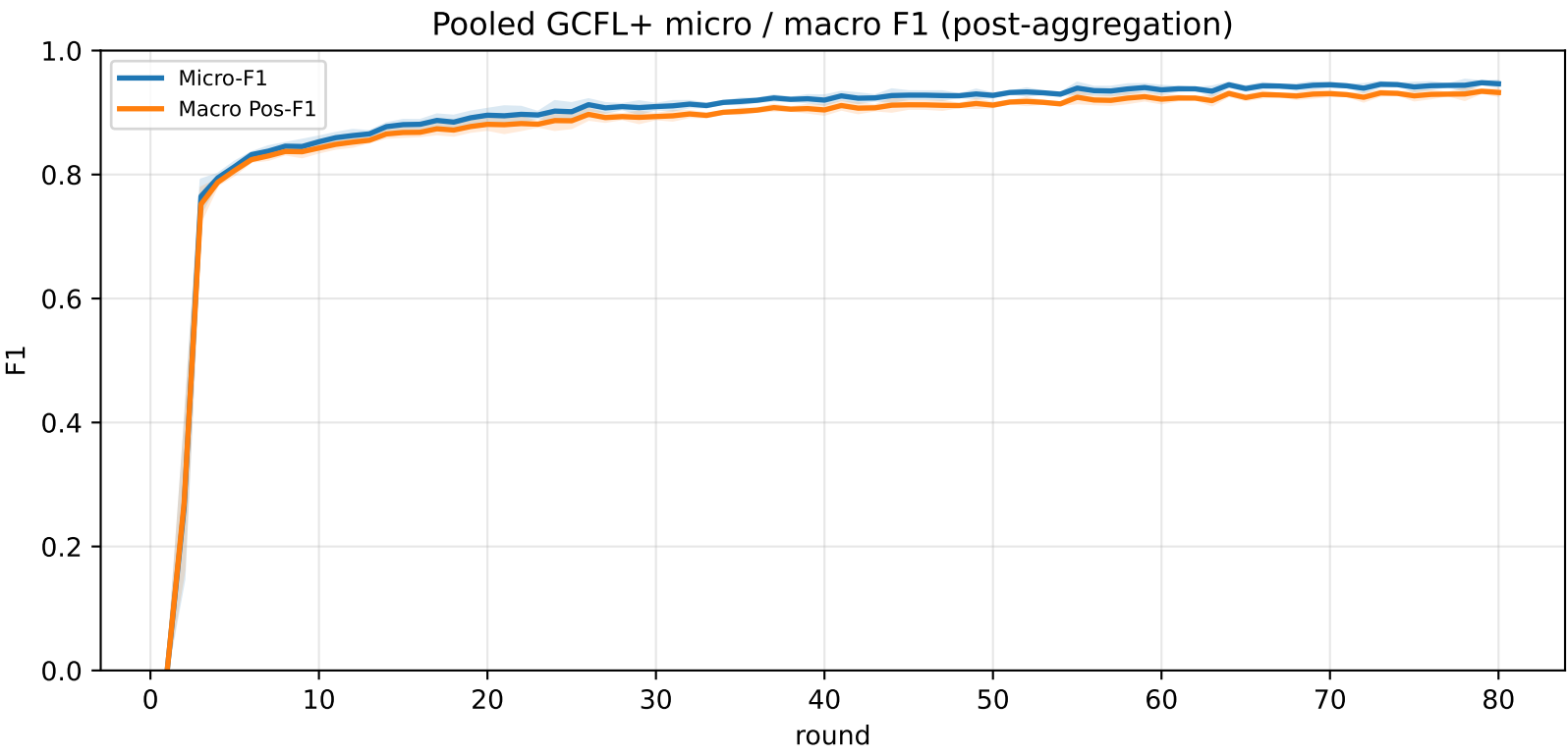
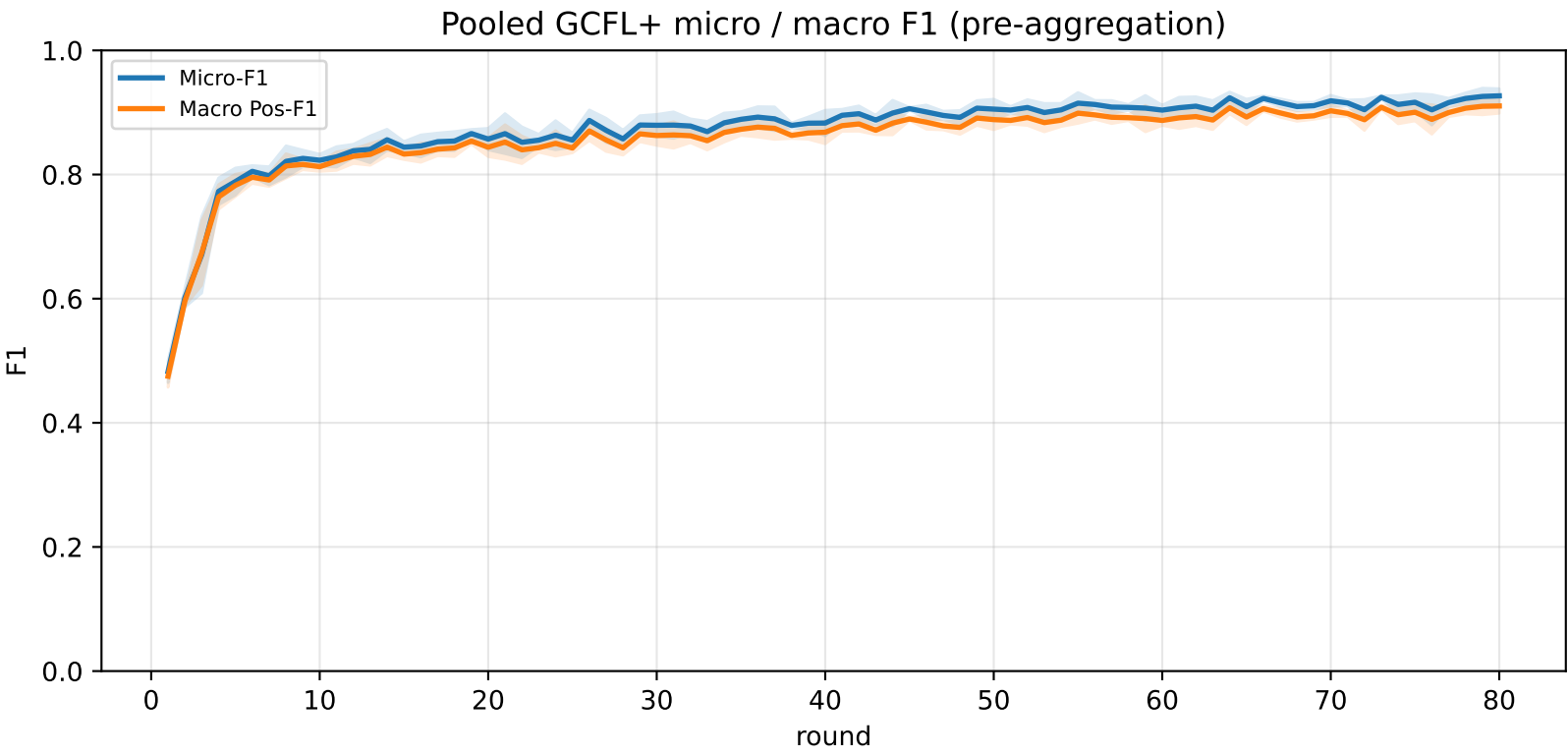
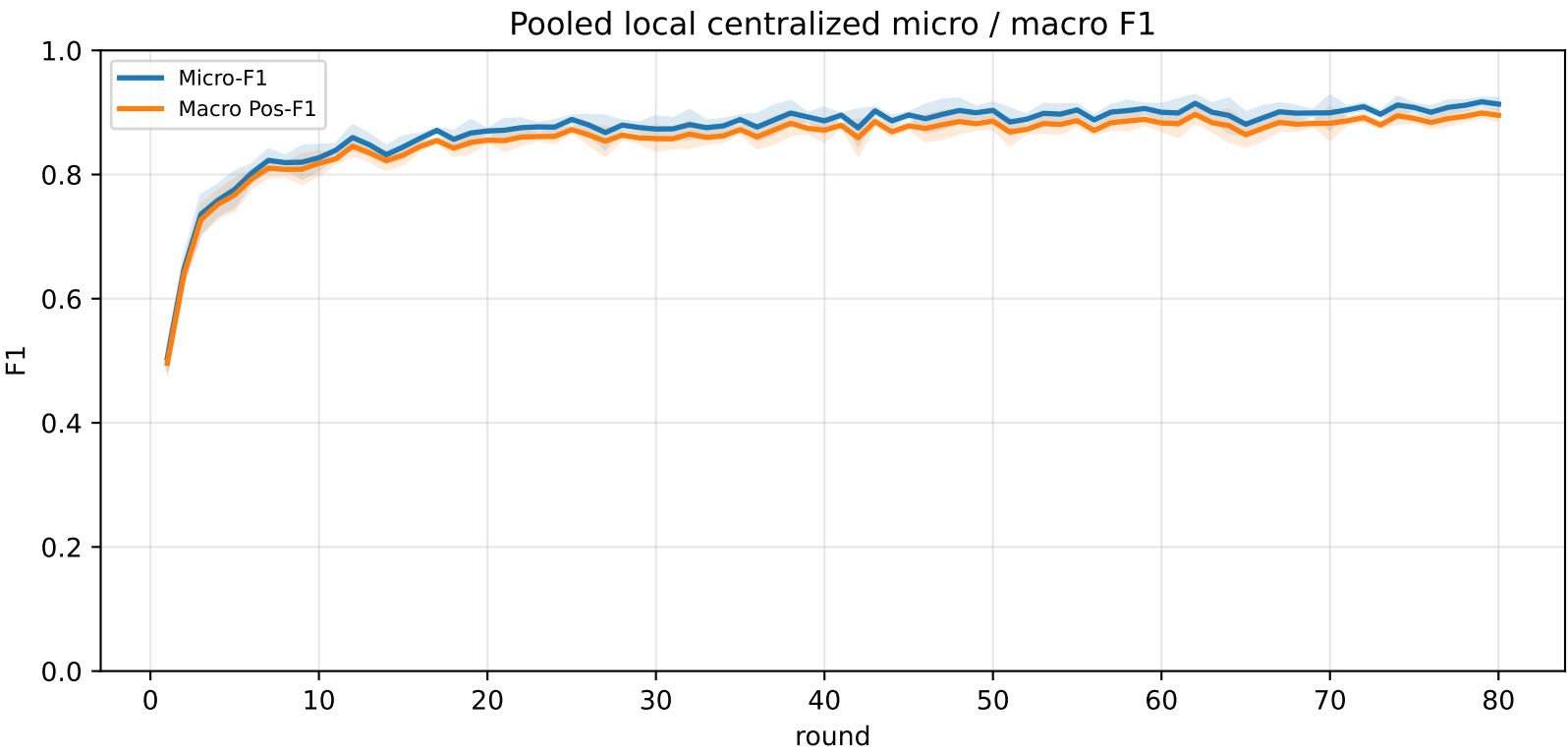
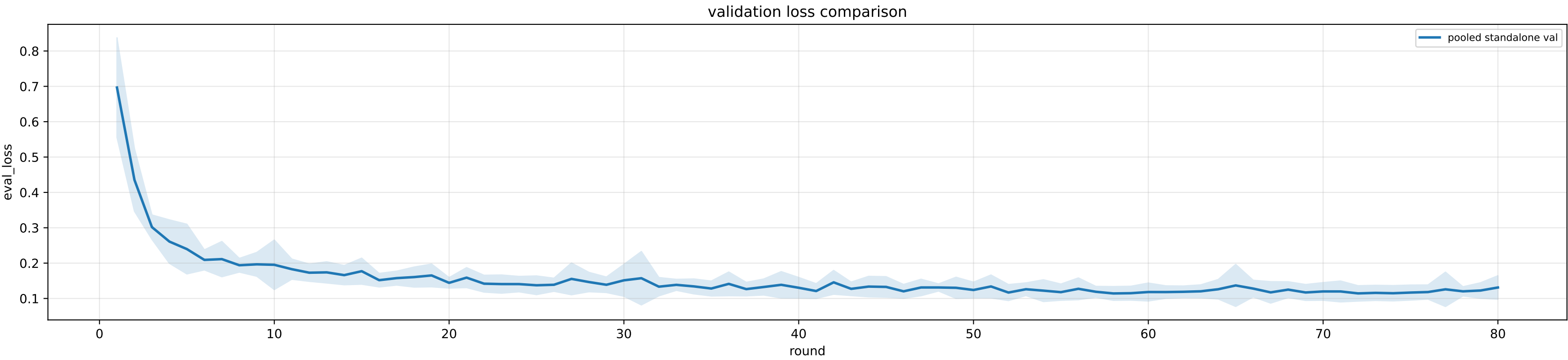
family=strong_decreasing | cycle2=37.8%±0.4% | cycle3=32.8%±0.8% | cycle4=27.9%±0.5% | cycle5=22.4%±0.5% | cycle6=16.0%±0.2%

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.5% ± 0.2%	92.5% ± 0.3%	99.6% ± 0.0%	99.1% ± 0.2%	96.8% ± 0.3%	90.2% ± 0.2%	76.6% ± 1.6%
GCFL+	92.9% ± 0.2%	90.8% ± 0.2%	99.6% ± 0.1%	99.2% ± 0.1%	95.8% ± 0.4%	89.1% ± 0.5%	70.8% ± 1.1%



family=mild_decreasing | cycle2=17.1%±0.7% | cycle3=14.6%±1.1% | cycle4=12.6%±0.5% | cycle5=9.5%±0.8% | cycle6=7.7%±0.4%

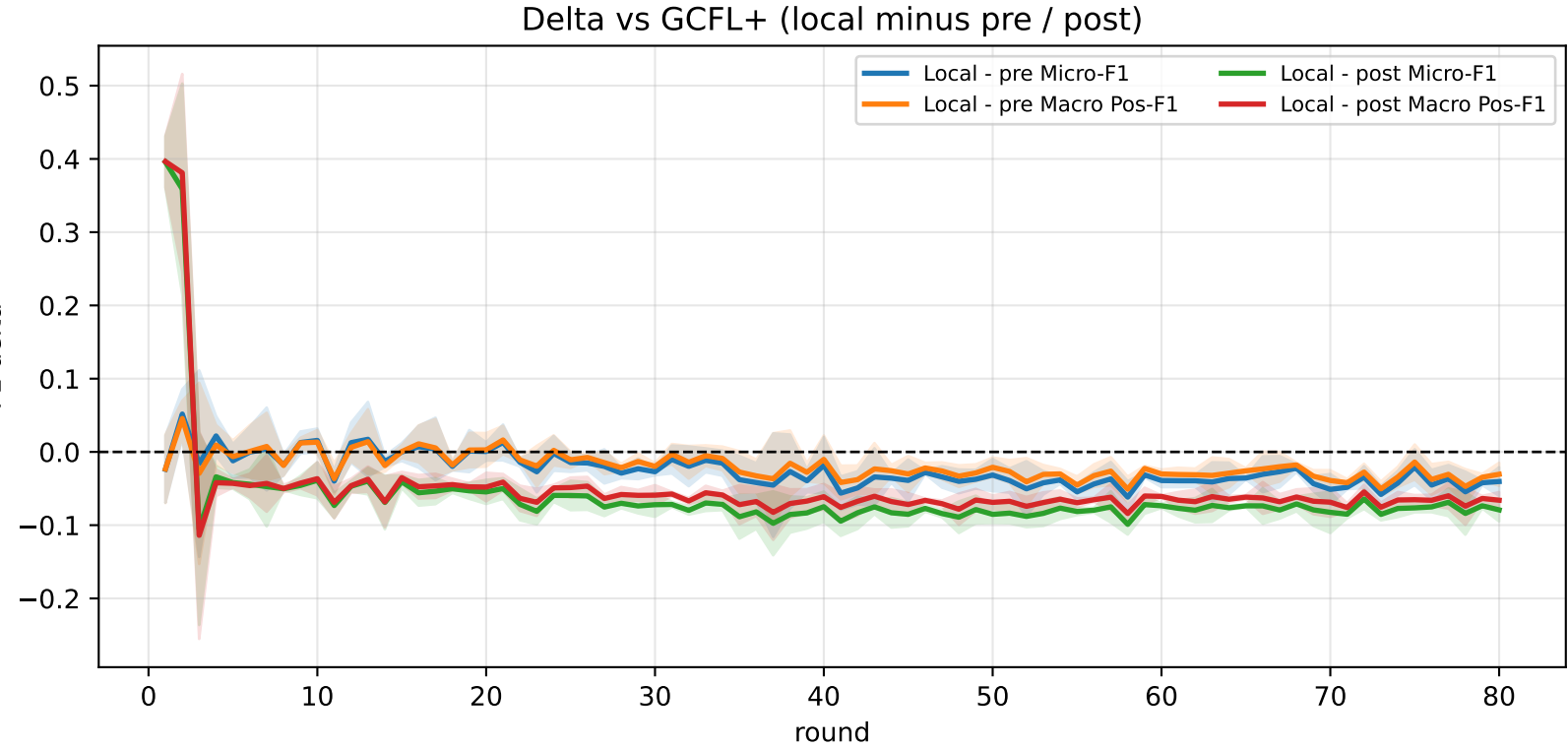
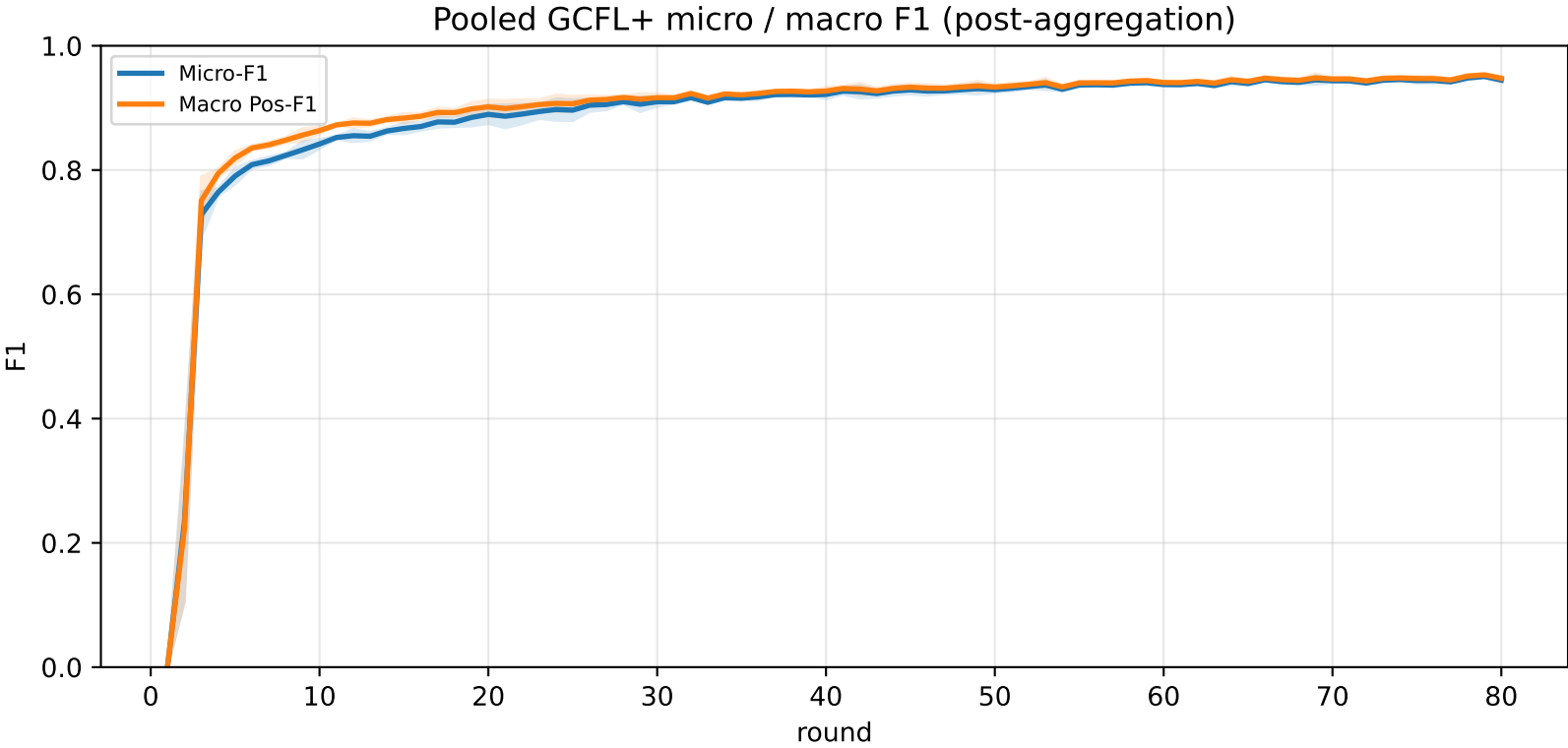
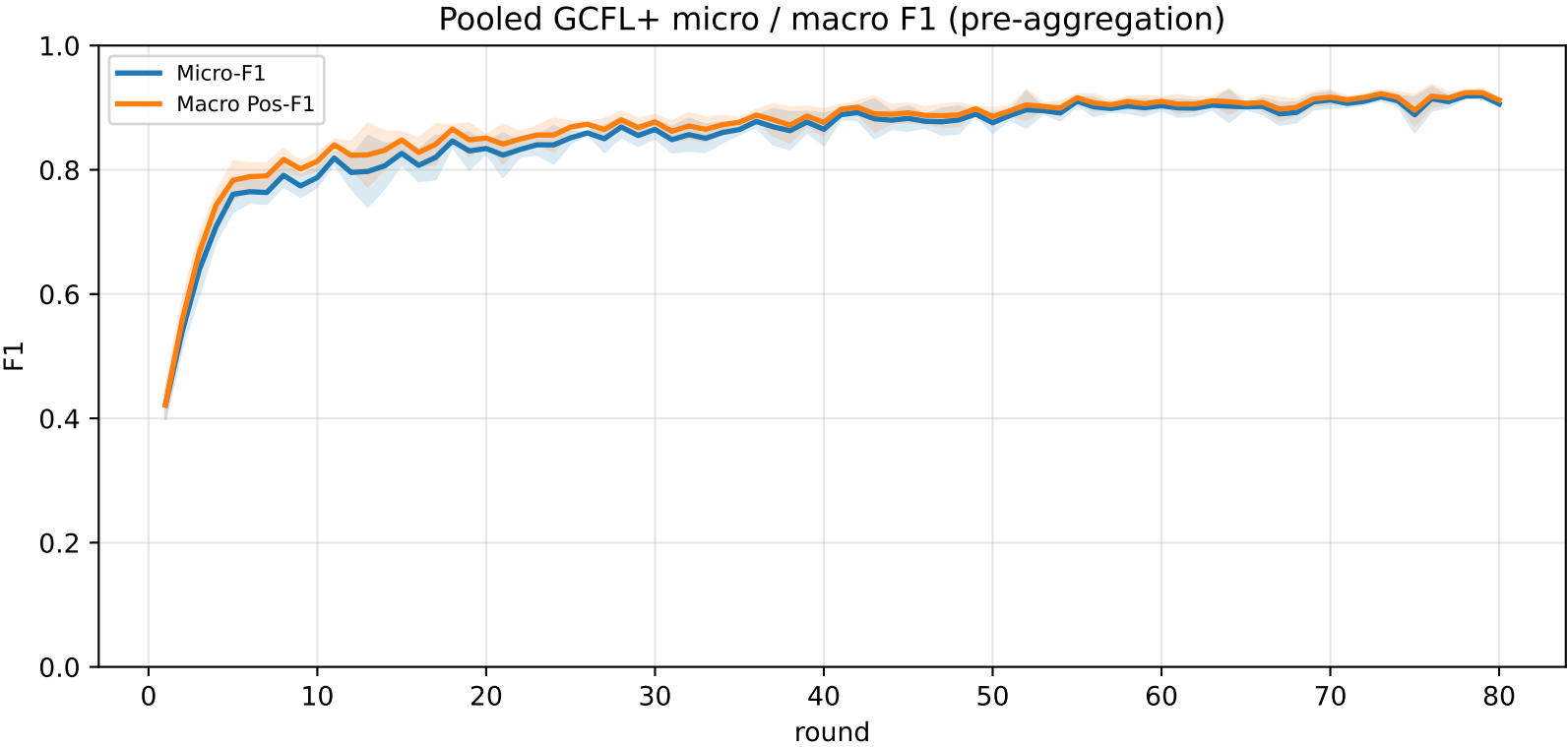
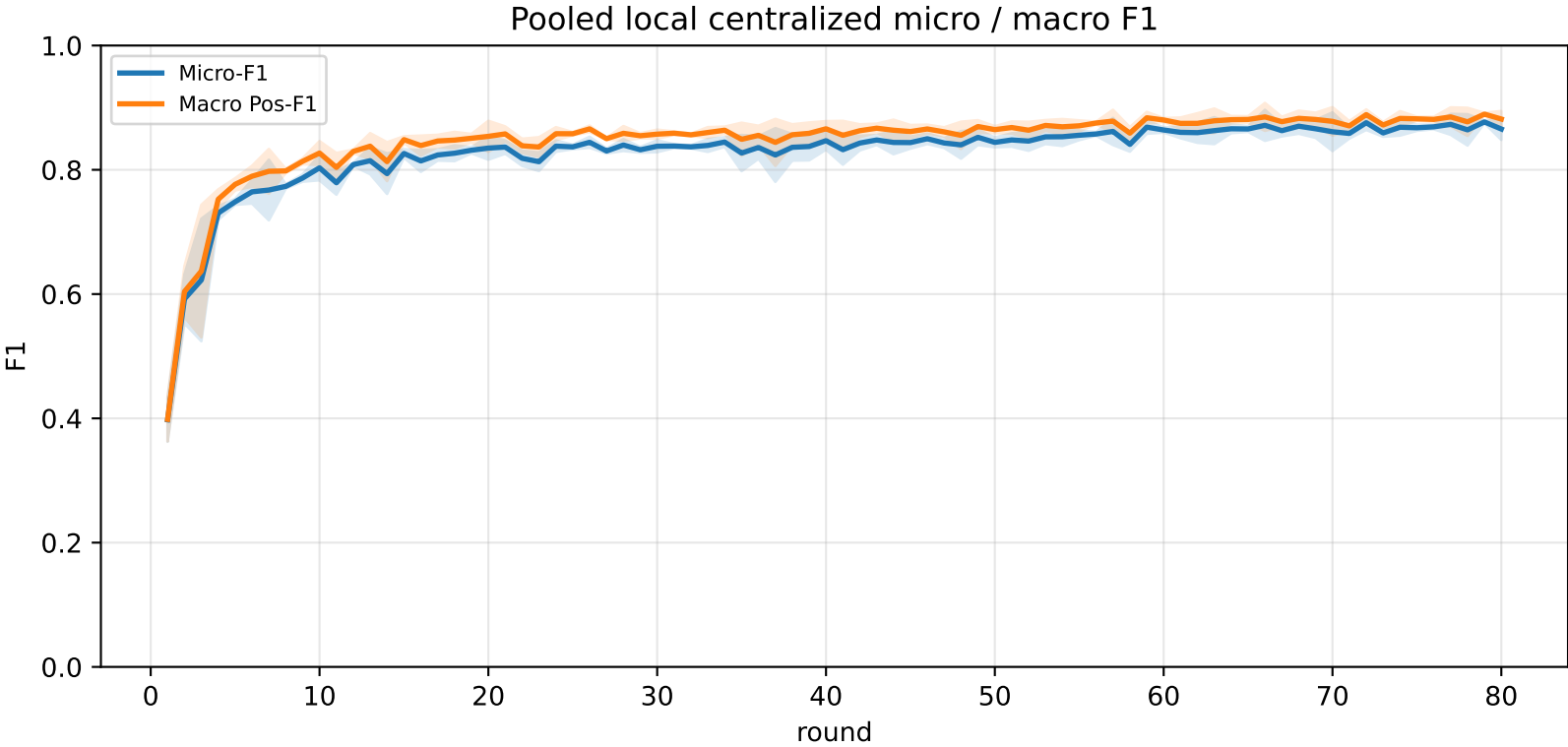
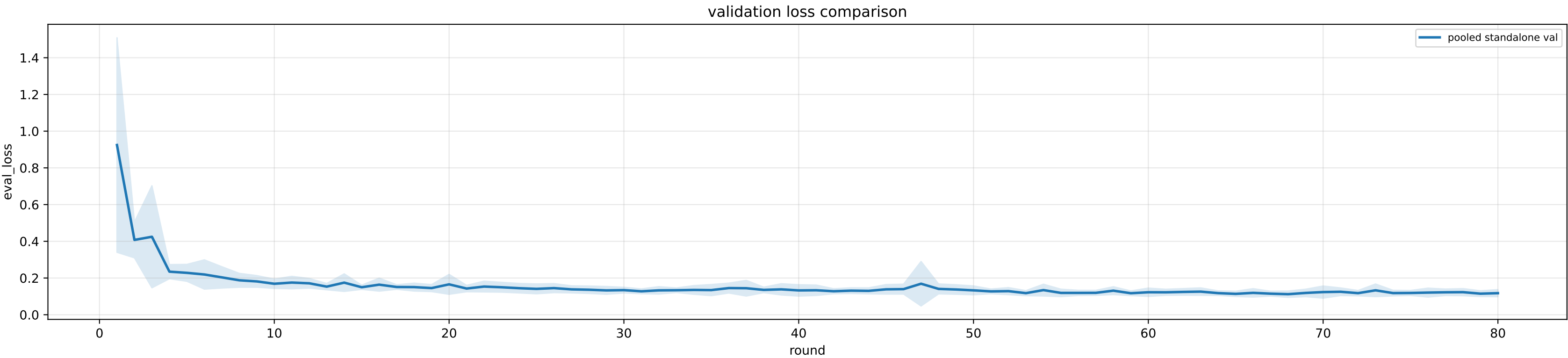
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	91.1% ± 1.1%	89.3% ± 1.1%	99.2% ± 0.1%	98.2% ± 0.1%	94.8% ± 0.6%	85.2% ± 2.2%	69.5% ± 4.1%
GCFL+	95.2% ± 0.2%	93.7% ± 0.2%	99.5% ± 0.1%	99.0% ± 0.1%	96.7% ± 0.1%	93.5% ± 0.3%	80.5% ± 1.1%



GCFL+ diagnostics | family=flat_balanced | n_clients=4 |
model=mcwauto_layers6_lr0.001_wd0.0001_do0.1_hd64_egoTrue_bs64

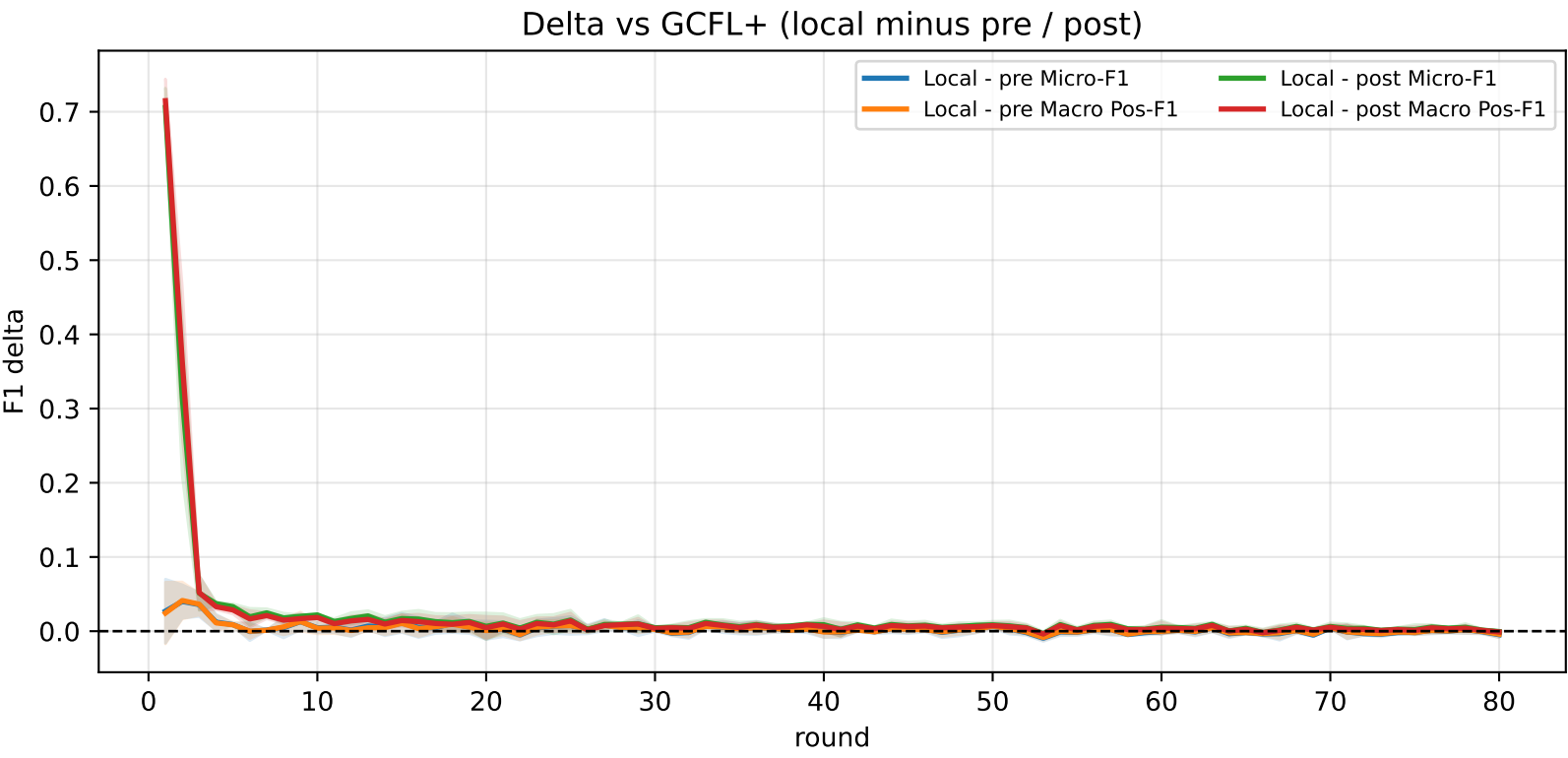
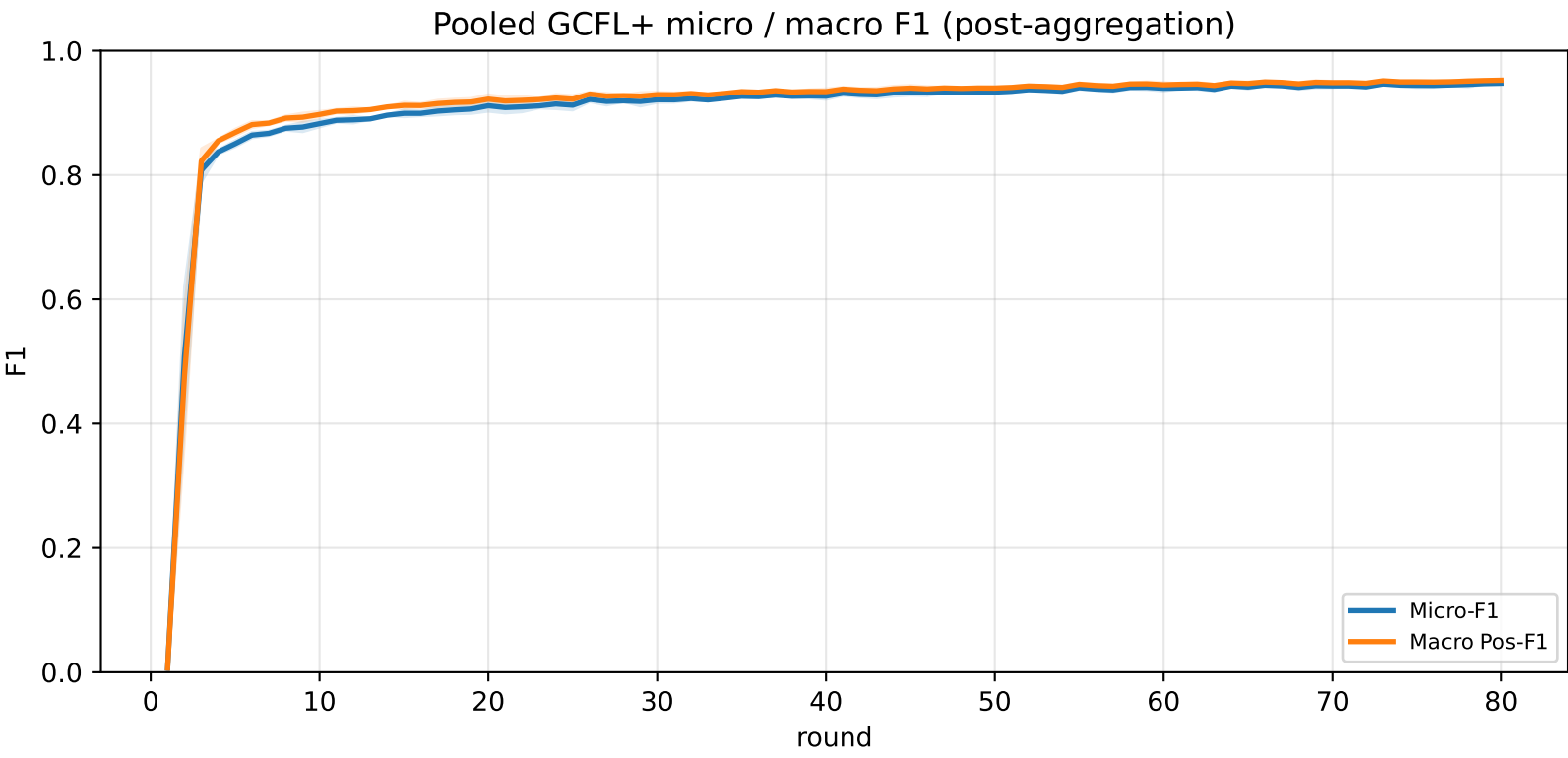
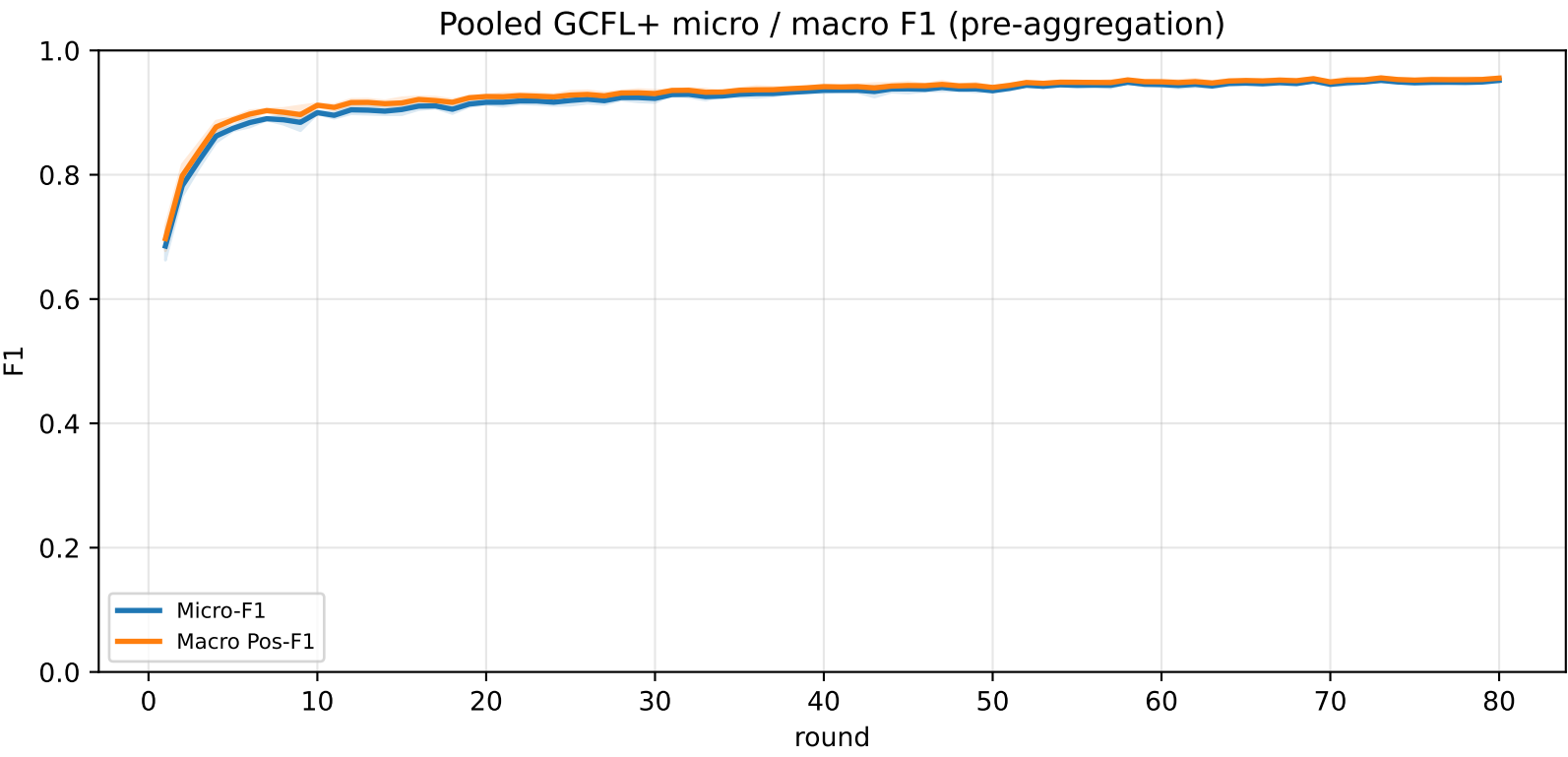
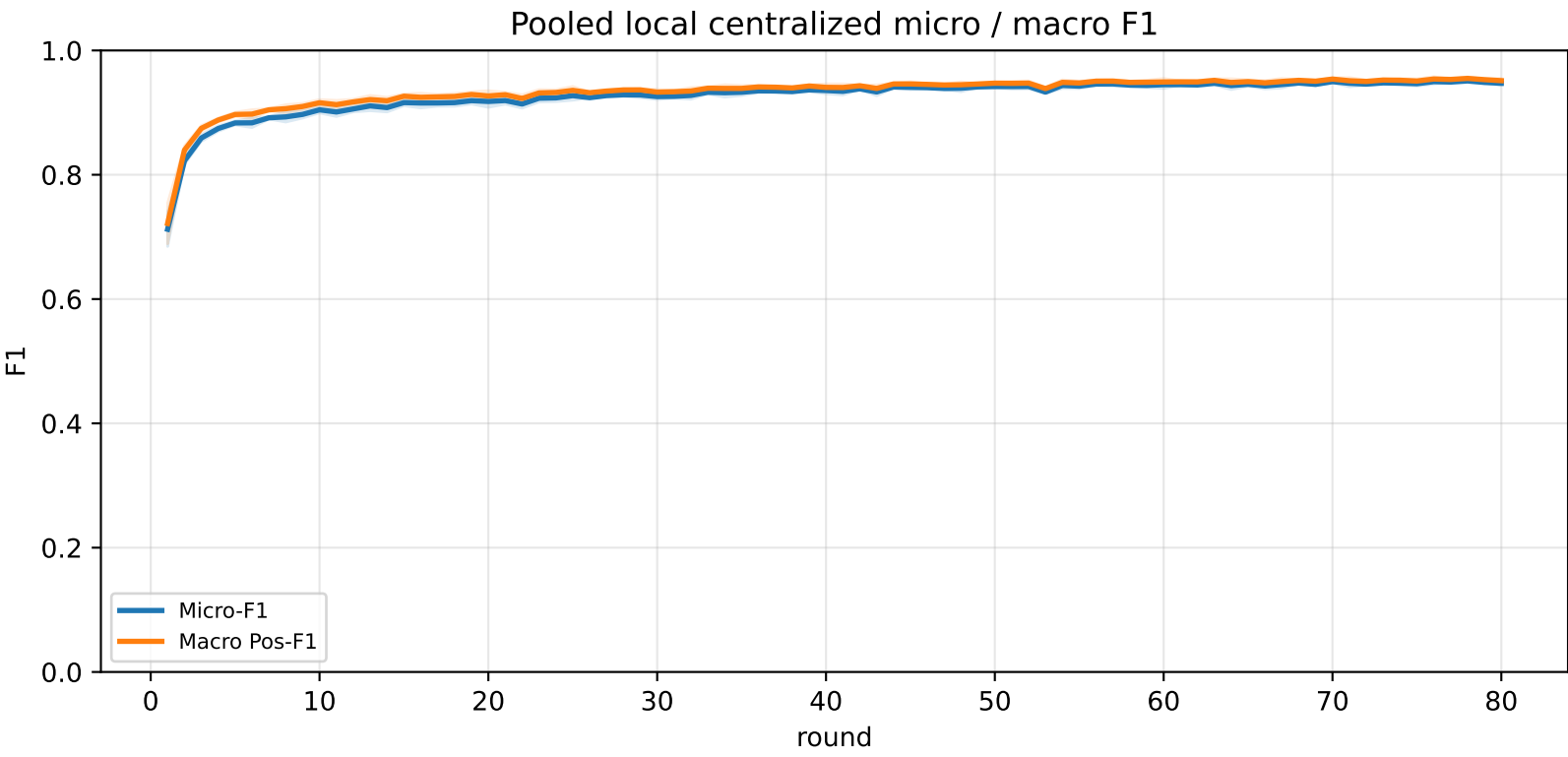
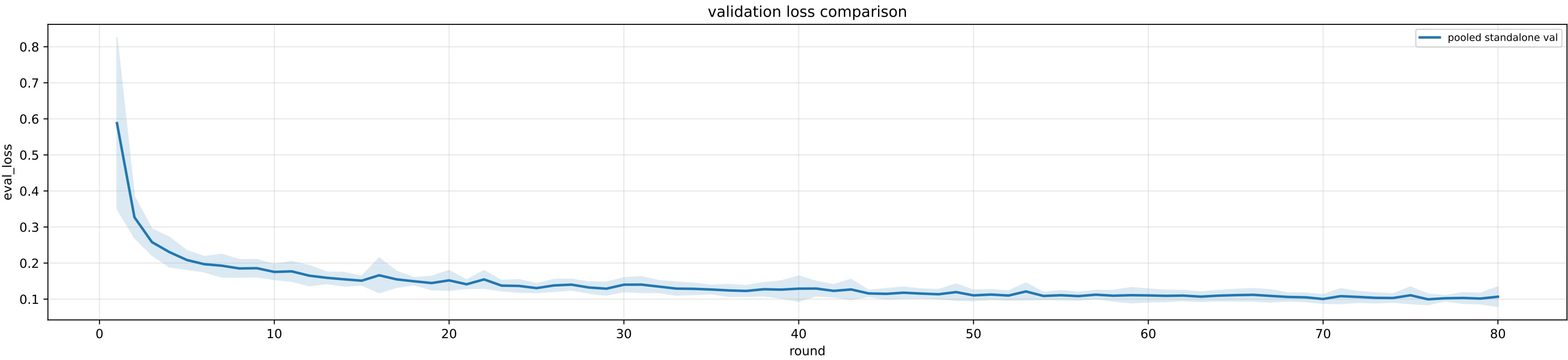
family=flat_balanced | cycle2=8.6%±0.7% | cycle3=10.2%±0.8% | cycle4=9.7%±1.5% | cycle5=9.3%±1.9% | cycle6=8.6%±0.7%

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	87.6% ± 0.2%	88.9% ± 0.1%	98.6% ± 0.2%	97.0% ± 0.1%	92.6% ± 0.7%	87.0% ± 0.8%	69.2% ± 0.5%
GCFL+	94.7% ± 0.3%	94.9% ± 0.3%	99.1% ± 0.2%	98.6% ± 0.4%	96.1% ± 0.7%	94.0% ± 0.3%	86.7% ± 0.7%



family=mild_increasing | cycle2=20.6%±0.6% | cycle3=24.0%±0.4% | cycle4=26.6%±0.7% | cycle5=29.1%±0.4% | cycle6=28.9%±0.9%

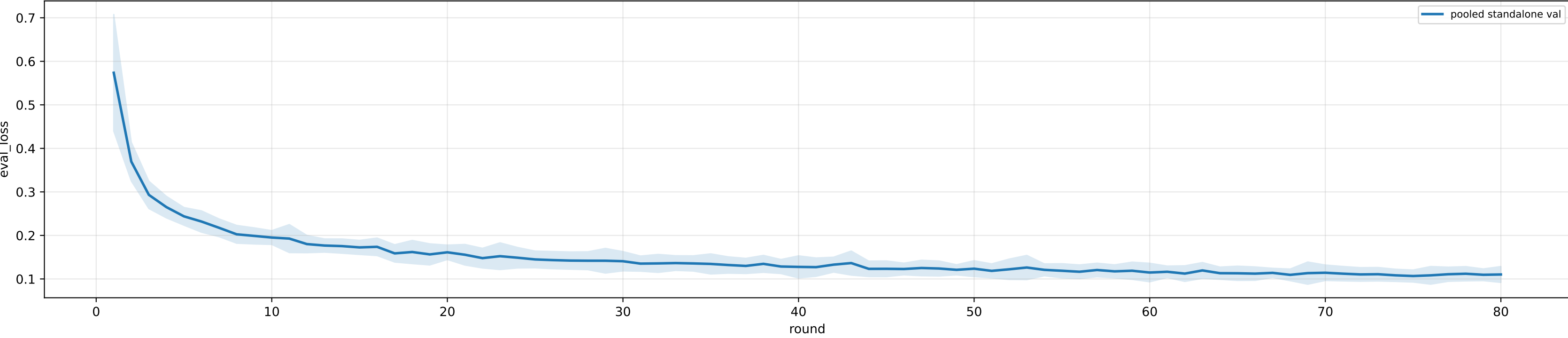
Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	94.9% ± 0.2%	95.3% ± 0.2%	99.0% ± 0.1%	98.4% ± 0.1%	97.0% ± 0.3%	93.8% ± 0.2%	88.4% ± 0.7%
GCFL+	94.8% ± 0.1%	95.2% ± 0.1%	98.8% ± 0.1%	98.3% ± 0.1%	95.8% ± 0.4%	94.2% ± 0.2%	88.8% ± 0.2%



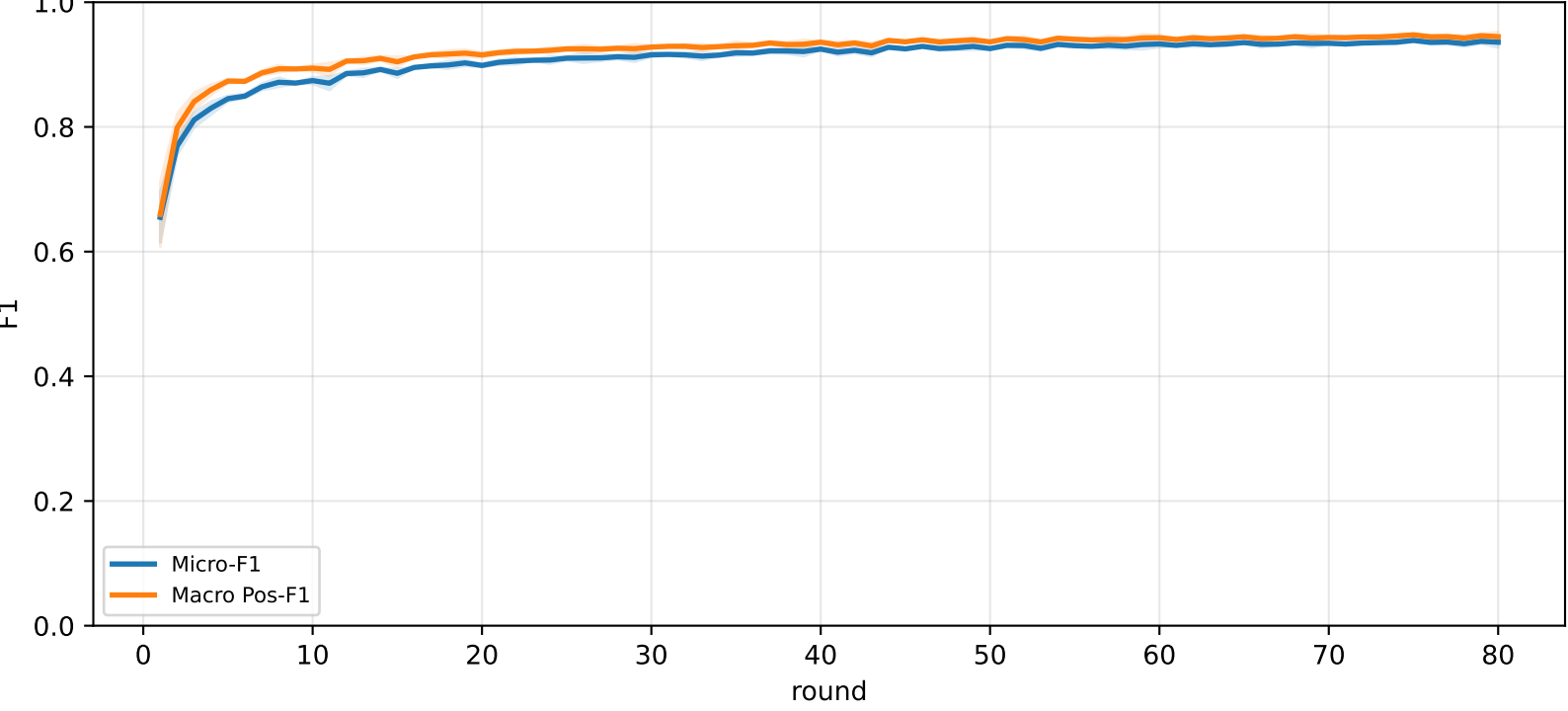
family=strong_increasing | cycle2=13.6%±0.4% | cycle3=18.0%±1.5% | cycle4=21.8%±0.4% | cycle5=26.1%±0.7% | cycle6=29.3%±0.4%

Algorithm	Micro-F1	Macro Pos-F1	cycle2 pos-F1	cycle3 pos-F1	cycle4 pos-F1	cycle5 pos-F1	cycle6 pos-F1
Local centralized	93.6% ± 0.2%	94.6% ± 0.1%	98.4% ± 0.2%	97.7% ± 0.1%	95.6% ± 0.1%	92.1% ± 0.6%	89.0% ± 0.4%
GCFL+	93.3% ± 0.3%	94.3% ± 0.3%	98.5% ± 0.1%	97.4% ± 0.1%	94.2% ± 0.9%	92.2% ± 0.4%	89.0% ± 0.3%

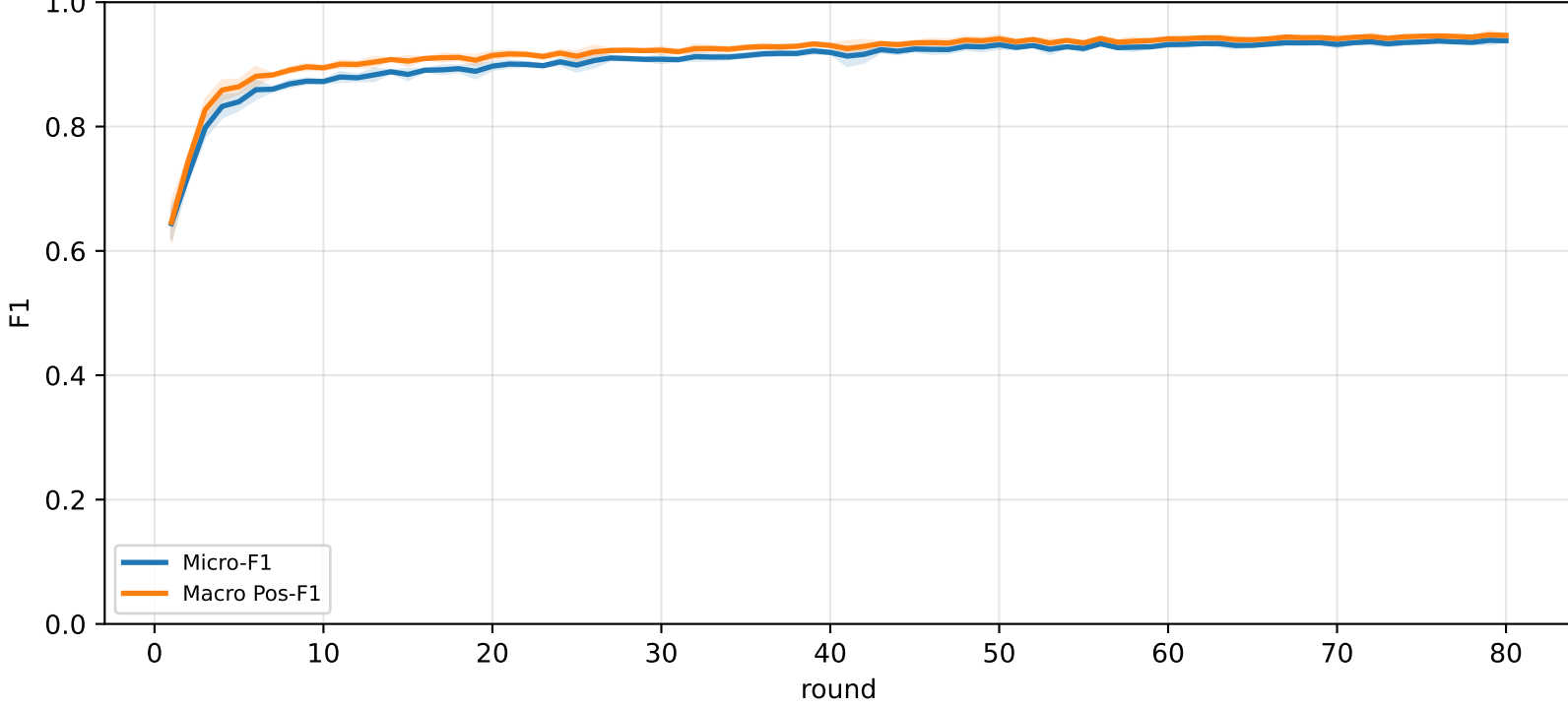
validation loss comparison



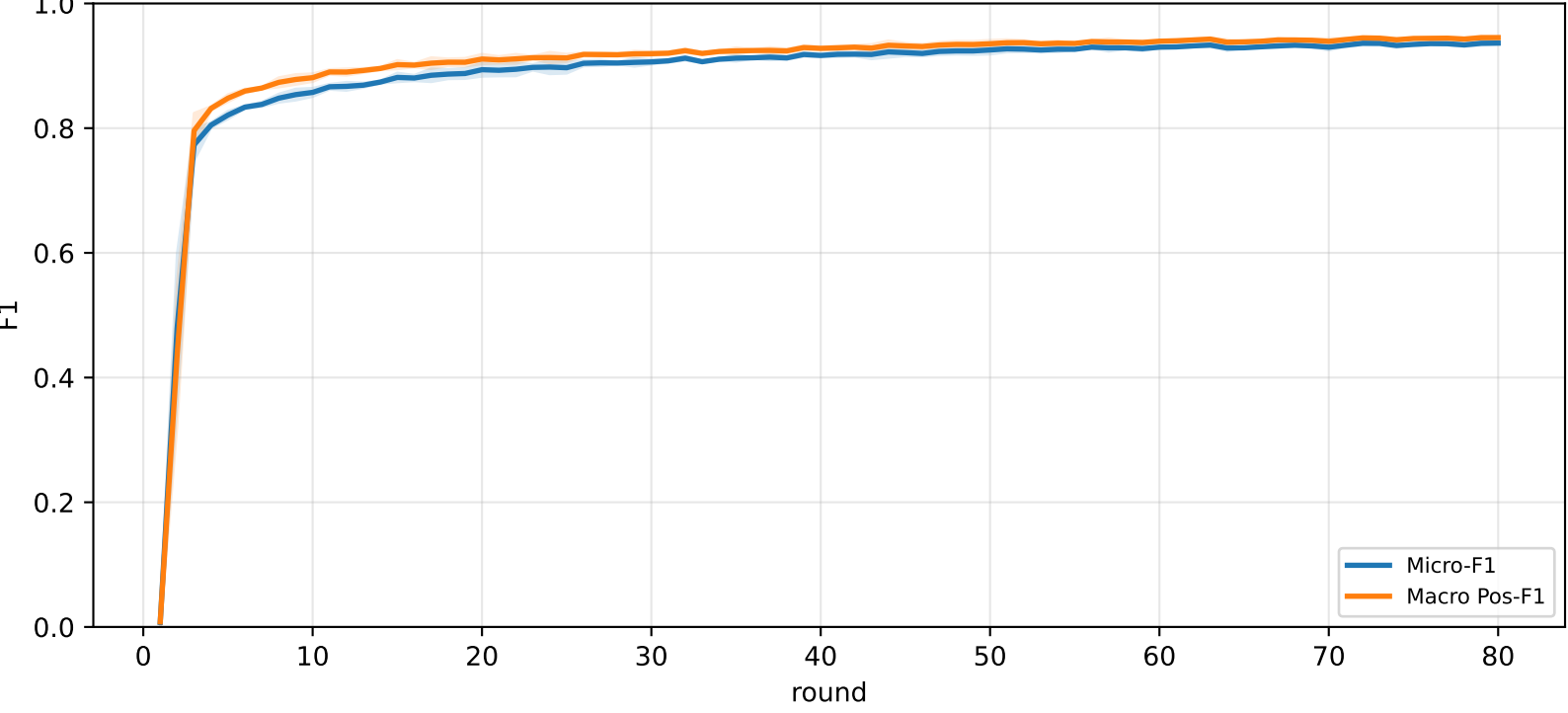
Pooled local centralized micro / macro F1



Pooled GCFL+ micro / macro F1 (pre-aggregation)



Pooled GCFL+ micro / macro F1 (post-aggregation)



Delta vs GCFL+ (local minus pre / post)

